Miscategorization as an Information Friction? Activating School-Based Agricultural Education in Liberia with Parental Engagement

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Abstract

Many developing countries have under-staffed agricultural extension systems, while rural households trade off children's schooling with farm labor. This paper shows schools can become cost-effective hubs tackling these dual challenges, but only if parent-school linkages are strengthened. In a three-year Liberian experiment across 197 schools, I evaluate school-based agricultural education (SBAE)—a voluntary program training science teachers to integrate inquiry-based learning with student-led school demonstration farms and home farming projects—estimating intent-to-treat effects on randomly sampled students. I cross-randomize whether to inform parents about this integrated program via promotional videos and field days. SBAE improves targeted outcomes when paired with parental engagement: students' households adopt 26–53% more yield-improving farming techniques, dropouts fall 4–5 percentage points per year, absenteeism falls 28%, and studying hours rise 22%. It is among the most cost-effective agricultural extension trials, and over twice as cost-effective for improving attendance as a conditional cash transfer benchmark. SBAE's positive training effects on teachers and students are unaltered by parental engagement. However, without parental engagement, SBAE's diffusion mechanisms—parents' learning from school farms, home projects, intra-household communication—stay muted. Technology adoption effects vanish, absenteeism rises 31%, studying hours fall 20%, and parents disconnect: only a third of parents visit school farms in three years, visit frequency falls 12%, and parent–teacher meetings fall 39%. Parents' persistent avoidance of information, plus sharply reversed responses under engagement, pose a puzzle. Bayesian updating, selective attention, and associative memory each fall short. I propose an integrative explanation: parental engagement shifts parents away from viewing agriculture and education as substitutes.

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1 Introduction

Rural sub-Saharan Africa faces two problems at once. Agricultural extension systems are severely understaffed, with ratios often approaching one extension officer per 5,000 farm households (Anderson and Feder, 2007; Moore, 2017), so farmers lack the technical knowledge to manage their soils, and agricultural productivity has barely grown since the 1960s (The World Bank, 2008).[1] At the same time, rural households trade off children's schooling against farm labor (Beegle et al., 2006; Edmonds, 2005). These two problems feed each other in a vicious cycle. Poorly managed soils degrade, and yields fall (Jayne et al., 2014; Sheahan and Barrett, 2017). Falling yields tighten household budgets, so parents put children to work on the farm, pull them out of school, and lower their aspirations for children's education (Bernard et al., 2014; Macours and Vakis, 2014). Children who leave school early become farmers without the human capital to try new technologies on their own plots and learn from the results (Foster and Rosenzweig, 1996; Wossen et al., 2017), and the educated workers who could staff extension systems never materialize (Juma, 2011; Bold et al., 2017). The cycle closes. Because each problem sustains the other, fixing one link is not enough: interventions that amplify social learning among farmers raise technology adoption (BenYishay and Mobarak, 2019; Munshi, 2004; Beaman et al., 2021), but have only weak immediate effects on children's schooling.

Breaking the cycle requires an intervention that tackles both problems at once. This paper studies an approach that turns schools from passive educational institutions into active catalysts for agricultural development. Specifically, I evaluate a school-based agricultural extension (SBAE) program that reaches farming households through their children while strengthening those children's engagement with school. Its education component trains teachers in experiential science instruction and engages students in extracurricular leadership and agribusiness training. Its extension component creates two complementary channels for technology diffusion: school farms serve as centralized demonstration plots for parents and nearby farmers, and students run home projects that function as decentralized demonstration plots on their family farms. This dual design lets farmers learn from socioeconomic peers, observe technologies under familiar conditions, and gather multiple observations before adopting, which matters especially in sub-Saharan Africa, where the performance of agricultural techniques varies greatly over small distances (Suri and Udry, 2022).

Despite its promise, SBAE faces two distinct risks in the fragile education systems typical of developing countries. The first risk is implementation inside schools. Teacher turnover and student dropout dissipate training investments; teachers may resist experiential instruction and the monitoring of student activities as diversions from exam preparation, or may lack the incentive to act as extension agents; and students may stay away if school farms are poorly maintained and offer few tangible benefits. On any of these counts, SBAE could prove less viable than traditional village-based extension, which reaches households through dedicated agents rather than through the education system.

The second risk, and the focus of this paper, lies outside schools: weak parent-school linkages. Even successful training of teachers and students may fail to change household farming practices and educational investments if parents are disconnected from the school. Parents in developing countries hold inaccurate beliefs about what and how much their children learn in school,[2] and these gaps persist because engagement is typically limited: schools rarely take the initiative to involve parents, parents with low educational aspirations see little value in visiting, and poor roads make the trip costly. Both of SBAE's diffusion channels require parents to be aware of the promoted techniques and of SBAE's student-centered training. Without that awareness, both channels can fail.

[1] This stands in stark contrast to Asia and Latin America, where Green Revolution technologies spread widely and raised yields (Gollin et al., 2021), and is all the more puzzling given decades of international development efforts aimed at African agriculture, including the establishment of international agricultural research centers in the 1970s (Evenson and Gollin, 2003), the Alliance for a Green Revolution in Africa (Denning et al., 2009), and substantial increases in aid flows to the sector (IFAD and UNEP, 2013).

[2] For example, Dizon-Ross (2019) shows that parents in Malawi hold wildly inaccurate beliefs about their children's academic performance even when report cards are available, with larger inaccuracies among higher socioeconomic status households.

Weak linkages pose a particular danger for students' education. SBAE offers households two things: (i) higher production through learning and adopting the promoted techniques; and (ii) student-centered training in agribusiness, better education, and access to a national network of mentors. If parents are unaware of (ii), SBAE can hurt schooling. Indeed, focus group discussions reveal that many parents confuse SBAE with typical school gardens, which use students as assistants. Such parents may simply view students as more skilled and obedient farm assistants, demand more child labor, cut school attendance, and keep educational aspirations low.[3]

This paper asks whether SBAE can overcome these barriers. It provides proof of concept that a developing country can build a cost-effective SBAE system that both diffuses yield-improving technologies to students' households and raises school attendance. Partnering with non-governmental organizations and government ministries, I randomly assign an SBAE program across 197 elementary and junior high schools in rural Liberia, where only 3.4% of farm households receive extension services (LISGIS, 2017) and 24% of students in the targeted grades drop out within two school years.

The program adapts the century-old 4-H model that brought practical agricultural education to rural America, and is designed so that students participate actively. Unlike typical school gardens in sub-Saharan Africa, where student labor is compulsory, participation is voluntary, and students rather than teachers make the major decisions on school farms, including which crops to grow and how to use the proceeds. The program pairs yield-improving technologies with agribusiness training and supports students in managing their own farms under teacher guidance, so that students can earn income. It is equally designed to strengthen school attendance and aspirations: teachers are trained to show how agriculture, science, and mathematics connect through experiential learning on school farms, which become open-air laboratories that partly compensate for the lack of textbooks, and a national network of annual agriculture fairs lets students compete, showcase their agribusiness success, and meet role models for future careers.

These design features address student and teacher engagement. Parental engagement remains the critical bottleneck: only 36% of program school parents attend parent-teacher meetings in a given school year, and only 44% report visiting school farms over the three-year trial. To address this bottleneck, I cross-randomize two light-touch parental engagement interventions within SBAE schools: (i) a one-time promotional video session, in which a parent of each sampled student watches a video explaining how the agricultural and educational components interact under SBAE, with testimonials from past participants and authority figures on how SBAE transforms students' ability to manage their own farms, their livelihoods, and their aspirations; and (ii) an annual farmer field day inviting parents to observe, discuss, and experiment with the promoted technologies on school farms. SBAE schools receive the video, the field day, both, or neither. Randomizing at the school level identifies SBAE's effects with and without parental engagement cleanly and accounts for spillovers within communities. Crucially, these interventions are cheap: together they account for only 10% of average per-school implementation costs, or 30% if the video is extended to all three cohorts of students in the trial period.

The evaluation mirrors the layered randomization. A follow-up survey of students in the first rainy season captures SBAE's effects before any parental engagement treatment. Midline surveys of students and endline surveys of students and parents in the second and third rainy seasons capture the effects of SBAE with and without engagement. I track 3,757 households, sampled from the pool of students with farming experience, which is highly representative of students in sampled schools, and estimate intent-to-treat effects throughout.

The results deliver one central message: parental engagement transforms what SBAE achieves. When SBAE is paired with both the video and the field days, the main yield-improving technique is adopted on 18 percentage points more students' parcels and 11 percentage points more parents' parcels, from control-school

[3] As I show in a companion paper (JMP2, Lee, 2025), weak parent-school linkages also amplify students' uncertainty about their parents' preferences and beliefs. Students hesitate to propose changes in farm management because they do not know whether parents are open to ceding control over resources, and so fail to make the timely earnings that could help them pay school fees.

rates of 29% and 74%. Considering the full package of soil management techniques, which have synergistic effects on yields, adoption on parents' and students' parcels rises by 26% (0.38 standard deviations) and 53% (0.48 standard deviations). These are large effects: traditional village-based extension programs in sub-Saharan Africa routinely fail to raise adoption among farmers who are not directly trained (Suri and Udry, 2022). The augmented program also substantially improves educational outcomes over two consecutive school years. Dropouts fall by an estimated 3.8 to 5 percentage points (from 22% at midline and 33% at endline), absenteeism falls by 18% and 28%, and weekly studying hours rise by 7% and 22%. Students also save more and show signs of aspiring to more education, without adverse side effects.[4]

Do these benefits justify the program's cost of $40 per student?[5] A back-of-the-envelope calculation under conservative assumptions says yes.[6] First, benefits outweigh costs in the three-year trial period, with a benefit-cost ratio of 1.33; the ratio stays close to one even if parents' gains from technology adoption are dropped entirely. Second, despite its broad scope, the augmented program is more cost-effective than major alternatives, judged by impact per $100 against other randomized trials in Africa. For technology adoption, it induces more households to adopt than every other rigorously evaluated village-based extension program except BenYishay and Mobarak (2019), which pays trained farmers to promote new technologies to others. For school attendance, it is 1.7 to 2.5 times as cost-effective as conditional cash transfers in Malawi (Baird et al., 2011). Third, the returns extend well beyond the trial. Once school-level capacity is built, monitoring costs fall sharply, so the program achieves a benefit-cost ratio of 1.53 over ten years even if annual benefits halve after year three. This robustness justifies long-term policy commitment: the institutional capacity and teacher training already in place can deliver sustained returns at a fraction of the initial cost.

What drives the pivotal role of parental engagement? A natural guess is that engagement raises students' responses to SBAE training. The evidence rejects this. The program has substantial training effects on its own: it improves teachers' knowledge of the promoted techniques and their use of experiential pedagogy, and the first follow-up survey, conducted before any parental engagement treatment, shows that students already use and know more about the promoted techniques, hold more accurate beliefs about attainable cassava yields, and are 9 percentage points more likely to intend to look after farms. By the endline, program participation reaches 81% without engagement, and SBAE alone generates large gains in students' agricultural knowledge and yield beliefs. Adding parental engagement improves none of these outcomes. If better responses to training explained the dramatic effects of engagement, we would expect positive marginal effects of engagement on these intermediate outcomes. We see none.

Instead, parental engagement activates SBAE's two diffusion channels, which otherwise stay completely muted. First, parents visit school farms and attend parent-teacher meetings. Without engagement, 81% of students in program schools have joined the program by the endline, yet only 44% of their parents report visiting the school farm, and program school parents are no more likely than control school parents to visit in the second rainy season. With engagement, 65% of parents visit, and visits in the second rainy season double. Relative to parents in SBAE-only schools, these parents report learning four times as many practices from schools, and 23 percentage points more of them have observed a harvest on the school farm. This ex-

[4] Students' spending and food consumption are unchanged, but 10 percentage points more students report positive savings, and Poisson estimates put the rise in average annual savings at 21% from a base of about $20, consistent with higher farm sales and parental transfers. The program raises the share of students taking agriculture as an elective by 18 percentage points, naming an agricultural career among their top three choices by 12 percentage points, and saving for university by 6 percentage points. Students shift time away from working as farm assistants while keeping total farming hours constant, contrary to the concern that SBAE would increase child labor. Future surveys will track crop yields on students' and parents' parcels and spillovers to non-parent farmers.

[5] This is the total cost per beneficiary across three cohorts, covering field officer and teacher training, school farm setup, school monitoring, and national events for extracurricular clubs.

[6] In valuing parents' gains from technology adoption, I count only the additional parents who adopt the main yield-improving technique, and assume they achieve only half of the yield gains observed in agronomic trials, on only a quarter of their cultivated area, and in only one season. In valuing students' discounted lifetime earnings, I assume a return of 8/200% per day of schooling, which implies an 8% annual return consistent with estimates of secondary school returns in sub-Saharan Africa.

posure translates into better knowledge (0.2 standard deviations) and beliefs about the promoted techniques.[7] Consistent with the reversal in school attendance, engagement also overturns SBAE's effect on parent-teacher meeting attendance, from a 39% decrease to a 37% increase.

Second, students manage their own farms and show parents the improved yields from the promoted techniques. Without engagement, the program does not raise the share of students involved in choosing farming methods on their own parcels, which stays indistinguishable from the control-school rate of 23%; this reflects the absence of home projects and mirrors the absence of adoption. With engagement, this share rises by 15.4 percentage points, the number of promoted techniques students show parents how to use rises by 92%, and, most importantly, 18 percentage points more parents report observing improved yields on students' parcels.

With both channels muted, the program fails completely without parental engagement. By the third rainy season, it does not raise adoption of the main yield-improving technique on students' or parents' parcels. Worse, the early warning in the first follow-up survey, where students spend 5% fewer hours per week studying, grows into clear harm to education: SBAE on its own raises absenteeism by 31% and cuts weekly studying hours by 20%.

Contrasting the two engagement treatments reveals what drives households' technology adoption and the reversal in educational decisions. Both treatments inform parents, but with different content. The field day delivers technical training on the promoted techniques, which parents can apply while keeping students as farm assistants. The video instead highlights the complementarity between students' agricultural work and their education under SBAE: students can manage their own farms and pursue further education that makes them better farmers. The results line up with this difference. Only the video, not the field day, raises the share of students who look after farms at midline. The video's effect on the number of techniques students show parents how to use is three times that of the field day, its effect on the number of techniques students advise parents to adopt is ten times as large, and the share of parents who observe improved cassava yields on students' parcels rises three times as much. Both treatments raise adoption on students' parcels, but only the video raises adoption on parents' parcels, which generate far more farm sales for households. Likewise, consistent with the video changing parents' aspirations for their children's education, the video accounts for the entire reduction in dropouts.[8] Parents who watch the video anticipate growth in students' farm management abilities, their beliefs about students' cassava yields remain significantly more optimistic at midline and endline, and beliefs measured immediately after the video strongly predict their subsequent school visits, ability to name field officers, and attendance at parent-teacher meetings.

These results pose a puzzle for standard accounts of belief formation. Without engagement, parents avoid cheap, readily available information: only 44% visit school farms over three years even though 81% of their children join the program, and their attendance at parent-teacher meetings falls. With engagement, a low-content promotional video reverses the sign of the program's effects. The reversal persists, extends from parents' engagement with the program to their children's schooling, and occurs even though parents in both arms face the same program and the same accessible evidence. In Section 7, I argue that Bayesian updating, selective attention (Schwartzstein, 2014; Hanna et al., 2014), and associative memory (Bordalo et al., 2020, 2026) each fall short of explaining these features jointly. I propose an integrative explanation: parental engagement shifts parents from viewing agriculture and education as substitutes to viewing them as complements. Without engagement, parents interpret SBAE through a familiar and widely shared view of school gardens as drudgery that trains obedient farm assistants. Under this view, a school that adds a farm is a school that teaches less. By showing how SBAE differs from typical school gardens, the video makes this view harder to hold. Parents become uncertain about which view is correct, and they start looking for evidence on the points where the two views disagree.

[7] Ten percentage points more parents believe students can produce more cassava tubers per plant than typical adults in their community.

[8] Disentangling the effects of different versions of the promotional video and subsequent engagement with students, Lee (2025) (JMP2) provides additional evidence that an interaction between parents' acceptance and students' requests for timely harvests of root crops, helping finance school fees, is crucial in explaining the video's effects on reduced dropouts under SBAE.

Related Literature. This paper provides proof of concept that schools in developing countries—particularly low-income African nations—can serve as cost-effective hubs for rural development by (i) diffusing improved agricultural technologies to students' households while (ii) increasing school attendance. In doing so, it deviates from a large literature proposing ways to strengthen existing agricultural extension systems that target adults as contact farmers, including providing incentives for contact farmers to communicate with peers (BenYishay and Mobarak, 2019), selecting contact farmers using social network data (Beaman et al., 2021), leveraging mobile phone advice to reach farmers directly (Aker, 2011; Fafchamps and Minten, 2012; Cole and Fernando, 2021; Fabregas et al., 2025), and targeting private input suppliers to disseminate information about new technologies (Dar et al., 2024). Most closely related are two papers examining the transmission of school-taught agricultural knowledge—through computer classes—from students to parents, using class-level randomization in Peru (Nakasone and Torero, 2016) and Uganda (Fernández et al., 2024). The Ugandan study, more relevant to the sub-Saharan African context, finds substantial gains in parents' agricultural knowledge but only modest increases in adoption of promoted techniques. Despite this promising evidence of student-to-parent knowledge transmission, important questions remain about the cost-effectiveness of promoting agricultural technology adoption in fragile education systems characterized by high teacher turnover and student dropout. This study addresses these gaps in four key respects. First, unlike the other two studies that rely solely on computer-based instruction, SBAE in the current study employs multiple channels for technology diffusion—including experiential science instruction, extracurricular activities, school demonstration farms, and student home projects. This multi-channel approach better addresses the barriers to social learning in African contexts, where vast heterogeneity in technology appropriateness poses unique challenges. Second, randomization occurs at the cluster-of-school rather than class level, with intent-to-treat effects accounting for voluntary student participation and potential dropout among both students and teachers. Third, we demonstrate that this integrative model is more cost-effective at increasing technology adoption than traditional extension approaches. Fourth, we show that school-based agricultural extension can be highly effective in regions where computer and internet infrastructure is absent—which represent the majority of rural areas in Africa.

This paper also contributes a new perspective to a large literature on the interconnections between students' agricultural work and education. Agricultural households in developing countries face a well-documented tradeoff between children's immediate value in production and long-term human capital investments.[9] At the same time, despite the fact that most students in rural Africa are projected to find employment in agriculture (Filmer and Fox, 2014), many are increasingly reluctant to pursue agricultural livelihoods due to perceived low returns and the view of farming as low-status work (Yeboah, 2018; Blimpo and Pugatch, 2021). This context leads to skepticism about school-based agricultural training programs for two reasons. First, such programs might intensify child labor and student absenteeism if they lead parents to view students as more obedient and effective farm assistants. Second, student participation in these programs might be low, and students might not take the training seriously if they aspire to non-agricultural careers. Against this backdrop, this paper documents a school-based agricultural training program that simultaneously (i) trains students effectively and increases their aspirations for agricultural careers; (ii) improves students' school attendance via reduced dropouts and absenteeism, as well as aspirations for progressing to universities; and (iii) increases students' experiences in managing agribusinesses without changing their overall working hours. More specifically, I find evidence that informational interventions persuading parents that students' agricultural work and education are compatible are crucial for (i), (ii), and (iii) to simultaneously happen.

Beyond immediate impacts, school-based agricultural extension can serve as a long-term development strategy. While the current RCT focuses on establishing proof-of-concept for technology adoption and attendance effects, the SBAE model's core components—student-centered training, agribusiness skills, and

[9] For example, households trade off children's farm labor with school attendance in response to enrollment subsidies (Ravallion and Wodon, 2000) and conditional cash transfers (De Janvry et al., 2006; Edmonds and Schady, 2012).

monthly student data collection—create opportunities for three synergistic interventions. These include: (i) providing experiential, student-centered pedagogy training for teachers who lack exposure to methods beyond traditional rote memorization (Nourani et al., 2025); (ii) enhancing agribusiness training by connecting school farm production to home-grown school feeding initiatives; and (iii) using high-frequency student outcome data to evaluate and refine which agricultural practices are most effective for dissemination. Together, these interventions transform SBAE into an enhanced platform that diffuses appropriate agricultural technologies while equipping students with generalizable skills beyond agriculture. This dual focus offers low-income countries a development strategy that simultaneously promotes rural livelihoods and prepares youth for structural transformation towards high-value jobs and sectors. I elaborate on these synergies, partnerships, and broader implications in the conclusion.

This paper also exemplifies how the hidden potential of school programs can be revealed through joint evaluations with household interventions. A large literature shows substantial room for improving educational outcomes through informational interventions that correct parental misperceptions, including beliefs about returns to education (Jensen, 2010), students' academic performance (Dizon-Ross, 2019), and study effort (Bergman, 2021). While these papers hint at weak parent-school linkages in developing countries, there have been no attempts to exploit this insight to enhance school program effectiveness. This paper, along with complementary findings in JMP2 (Lee, 2025), demonstrates that strengthening parent-school linkages delivers exceptional value for both agricultural technology diffusion and rural education through SBAE. Remarkably, while parental engagement proves pivotal to program success, it costs only less than 30% of SBAE's total budget. Furthermore, the strong evidence that parental engagement activates SBAE's mechanisms suggests that evaluations of school programs should routinely test parental engagement components to avoid underestimating program impacts. This insight has broader implications for program evaluation, as demonstrated success is typically necessary to generate buy-in for subsequent trials and scaling efforts, potentially biasing meta-analyses toward programs evaluated without key complementary components.

Finally, the puzzle posed by parents' responses contributes to the literature on belief formation. Neither Bayesian updating, selective attention (Schwartzstein, 2014; Hanna et al., 2014), nor associative memory (Bordalo et al., 2020, 2022, 2026) on its own explains why parents avoid cheap information about a program and then change course after a low-content video. Instead, the evidence suggests that parental engagement changes how parents relate agriculture and education, from viewing them as substitutes to viewing them as complements. This mechanism differs from the information frictions about returns to education or children's performance studied in earlier work (Jensen, 2010; Dizon-Ross, 2019; Bergman, 2021).

Outline. The rest of this paper is structured as follows. Section 2 provides the study background and introduces the school-based agricultural extension program in Liberia. Section 3 explains the experimental design. Section 4 presents the main results: the effects of SBAE with and without parental engagement, and the marginal effects of the promotional video and the farmer field day. Section 5 provides cost-benefit and cost-effectiveness analyses of the SBAE program when the promotional video and the farmer field day are included. Section 6 examines mechanisms: SBAE's training effects on teachers and students, which do not depend on parental engagement, and the diffusion mechanisms that stay muted without it. Section 7 sets out the puzzle posed by parents' responses and discusses its interpretation. Section 8 concludes.

2 Study background

2.1 Agriculture and education in Liberia

This study is conducted in Liberia where 72% of households live in absolute poverty (Benfica et al., 2018). Agriculture is the primary source of livelihood for about 80% of the population (African Development Bank, 2008). Smallholder farmers in this tropical region traditionally employ a fallow system under collective

ownership of the land.[10] This system operates smoothly with minimal inputs to the soil — no tillage nor use of fertilizers — if the fallow period is sufficiently long. However, as in many other sub-Saharan African countries, farmers are forced to shorten the fallow period under economic and population pressures, leading to commonly observed soil degradation and declining agricultural output.

Agricultural intensification — increasing output per unit of land by increasing inputs to the soil — is the common response to soil degradation. Managing the shift to agricultural intensification, however, requires farmers to acquire soil management techniques that are fundamentally different from traditional techniques. In Liberia, this progress is hindered by low levels of coverage by extension services (3.4%) and low levels of education — the average household head has four years of schooling. As a result, Liberia has very low levels of yields in major staples even by the standard of sub-Saharan Africa — FAO (2023) data suggests that rice and cassava yields are only 54% and 62% of the regional average.

Besides low agricultural productivity, Liberia also has an education system with lower school attendance than those in most sub-Saharan African nations. Liberia has a net primary enrollment rate of only 44% in 2017, compared to the regional average of about 80% in sub-Saharan Africa (UNESCO Institute for Statistics, 2020). Overage enrollment is particularly serious — over 40% of primary school students are above 12 years old; and about 50% of junior high students are 18-20 years old (Liberian Ministry of Education, 2017). Another reason is the high annual rates of student dropouts, which stand at 10-15% in upper primary and junior high grades.

Other than socioeconomic factors, such as poverty and needs for farm labor, a lack of quality instruction and educational resources are important reasons discouraging school attendance. Only 51% of all teachers are qualified to teach — this drops to 33% in junior high. Lacking the required certificates, unqualified teachers also do not receive in-service teacher training from the government. A lack of essential educational resources, such as textbooks, further worsen teacher instruction. Many students are forced to spend a significant portion of class time on copying from the blackboard. These factors are particularly discouraging for rural students who anticipate an agricultural career, as the way knowledge is taught hardly feels relevant to them.

2.2 School-based agricultural extension as an integrative model

School-based agricultural extension (SBAE) has a century-old history in rural America, dating back to the Smith-Hughes Act of 1917. At its core is an integrative model that blends the promotion of modern farming practices with student-centered learning experiences. According to Croom (2008), this model has three interrelated components. Through formal instruction, students learn the scientific basis of modern farming practices through lectures and demonstration. But more important are the other two components that ensure personalized learning experiences for rural students. Through participation in extra-curricular agricultural clubs, students develop their sense of belonging in a national network, meet mentors and role models, and gain leadership experiences and recognition in public services and competitions. Through supervised agricultural experiences, students can develop home projects tailored to their own pace and needs. Rufus Stimson, an early advocate of the method, argues in particular that home projects allow students to profit from farm work and complement school projects that are subject to overcrowding amongst students (Stimson, 1919).

In the sub-Saharan African context, SBAE is an important new model in the delivery of agricultural extension and rural education.[11] In particular, SBAE should be distinguished from typical school gardens due to its emphasis on student-centered learning experiences. Typical school gardens are directed by teachers without integration with scientific instruction of modern farming practices, home projects, and a national

[10] I use the term collective ownership here to refer to communal land tenure or lineage-based land tenure.

[11] As in the case of agricultural education in early US history, separate provisions of agricultural training and agricultural youth clubs can be found in some sub-Saharan African countries — for example, agricultural youth clubs affiliated with the international 4-H network can be found in 15 African countries. But the integrative SBAE model has only been introduced by the US-based NGO AgriCorps to Liberia and Ghana since the 2010s.

network of agricultural clubs. By contrast, students participate in SBAE voluntarily, make major decisions on school farms (e.g. choice of crops and use of proceeds), and are empowered to make profits from home projects and gain social recognition from clubs. These distinguishing features of SBAE make it particularly suited to tackle intertwined challenges in agricultural extension and school attendance.

2.3 Components of the 4-H Liberia program

This study examines an SBAE program in Liberia that is conducted by the local NGO 4-H Liberia in collaboration with the ministries of agriculture and education. Focusing on the RCT period from January 2021 to December 2023, this program implements the three-component model mainly through in-service teacher training and regular extension visits by extension officers.

Teacher training is offered twice a year to two teachers per school. These teachers are selected by the participating schools with the requirement that they teach either science or agriculture. The duration of each training workshop is 20 hours that are spread across several days, with roughly equal time covering (i) the integrative SBAE model; and (ii) the promoted yield-improving agricultural technologies. To ensure proper implementation of the SBAE model and diffusion of technologies, teacher training is reinforced by monthly extension visits. During these visits, extension officers monitor progress on school farms, collect information from student participants, and provide lessons on agriculture.

The 4-H Liberia program implements the three-component SBAE model in the following ways. First, formal instruction of agriculture and science is offered through a combination of extension visits and experiential instruction by teachers. Students attend lessons provided by extension officers, while teachers are given instructional ideas that they can adopt during classes. In particular, teachers are given structured lesson plans to illustrate scientific concepts that are associated with promoted farming practices, such as (i) using seed germination tests to illustrate how to conduct a scientific experiment; (ii) using farm site selection to illustrate what a soil profile is; and (iii) using the practice of composting to illustrate the concept of a chemical compound. Besides work on the school farm, teachers are also encouraged to use other experiential methods such as field trips, role plays, and dramas to pique students' interest in scientific knowledge.

Then, extra-curricular 4-H Clubs provide a series of learning experiences. In each school year, every participating school recruits 25-30 club members. These members then participate in, on average, two hours of farm work per week after school. They also meet, on average, on a bi-weekly basis to discuss club affairs. The most important issues are about the choice of crops in a growing season, ways to guard the school farm against thieves and pests, and ways to use the proceeds from farm. By enrolling in the club, students also earn the opportunity to represent their schools in annual national events. The most important event is the national agriculture fair, during which students compete in displaying their knowledge of agriculture to judges and share stories of successes in applying modern farming practices.

Lastly, the program aims to amplify the diffusion of agricultural technologies to rural communities through (i) using the school farm as a demonstration plot; and (ii) encouraging students to start home projects. The school farm is instructed to be open to visitors, with posters encouraging learning of promoted techniques by parents and farmers. To enable students to manage their own farms and make profits from the adoption of promoted techniques, extension officers provide lessons on agribusiness, including farm management and record keeping practices. Teachers are also encouraged to supervise students through home and farm visits.

2.4 Promoted agricultural technologies

Agriculture is an inherently complex activity that requires flexible practices with respect to local environments. The 4-H Liberia program thus provides a wide range of techniques that extension officers and teachers can choose to implement on the school farm. These techniques are grouped under the categories "soil management" (e.g. bed preparation), "crop protection" (e.g. weeding), "improved input use" (e.g. improved seeds or

crop varieties), "improved planting practices" (e.g. transplanting), "improved nursery practices", "harvesting practices", and "record keeping practices".

Under the advice of the Liberian Central Agricultural Research Institute, we focus on four soil management techniques that are most widely adopted and expected to improve crop yields, based on agronomic trials and findings from agricultural development programs in Liberia. The four specific techniques are (i) bed preparation methods; (ii) composting; (iii) regular spacing; and (iv) row planting.

These four techniques are promoted to improve agricultural productivity amidst soil degradation in a bush fallow system. Farmers employing the system typically clear land by cutting and burning vegetation. Because of the fertile soils created, farmers do not need extra tillage and instead sow seeds directly into the ground. To reduce the risk of single crop failures and pests, farmers often plant several crops, such as rice and cassava, in proximity. While these practices quickly drain soil nutrients, they are sustainable with traditional fallow periods, which could be as long as 10-15 years. With reduced fallow periods, however, farmers are increasingly unable to cope with agronomic problems such as soil erosion, soil compaction, and waterlogging.

Bed preparation methods, which include mounds or ridges for root crops and raised beds for vegetables, are the main yield-improving technique in this context. These methods help loosen the soil and enhance drainage, thus facilitating root penetration while avoiding waterlogging and topsoil erosion. Other promoted techniques tend to reinforce the effects of beds — with larger yield gains when adopted together. Composting provides a low-cost alternative to chemical fertilizers for improving soil fertility. Row planting and regular crop spacing help optimize the growth of individual plants by avoiding inter-plant competition for nutrients.

Agronomic trials in the tropical humid region in general recommend the promotion of these soil management techniques. In particular, several agronomic trials by the International Institute of Tropical Agriculture (IITA) suggest that the use of mounds and ridges can significantly increase root crop yields — cassava by 31-46% (Odjugo, 2008) and sweet potatoes by 70% (Agbede and Adekiya, 2009). These trials have two important features. First, the yield gains are found under conditions that mimic the traditional cultivation style with minimal soil inputs and random spacing. Second, consistent with anecdotes from program participants, the differences are easily visible, as indicated in the stem height and leaf size within 3 months of planting, and the number and size of tubers per plant at harvest.

These techniques are widely applicable for root crops and vegetables, which are grown by over 80% and 60% of households respectively (Liberia Institute of Statistics and Geo-Information Services (LISGIS), 2017). Farmers are also generally aware of these techniques — for example, 80% of control school parents are able to describe mounds or ridges as a way of planting root crops and 74% apply bed preparation methods to some parts of their farms in our endline survey.

However, a mixture of risks and increased labor can deter adoption by rural households. Though beneficial on average, these techniques tend to perform differently under heterogeneous growing conditions. Unsure about the benefits, farmers might not incur the additional labor for bed preparation — in the appendix to Section 5, I estimate that the labor takes about 150 hours per hectare in a season. In addition, while these techniques are particularly beneficial for root crops, farmers might be reluctant to change their practices due to to the important role root crops play in maintaining subsistence. For example, cassava tubers can be stored underground for up to 24 months while its leaves are consumed. One drawback of using bed preparation methods, which loosen the soil, is that farmers might be forced to harvest all produce together at once.

Because of these concerns, farmers seldom apply these techniques at a large scale and rarely apply all techniques together. For example, farmers commonly construct mounds opportunistically (e.g. where ashes are concentrated or where soils are depleted) and plant multiple cassava stakes, along with other crops, in and along the edges of each mound — minimizing labor and insuring against sprouting failures while missing the benefits from consistent spacing. The proportion of control school parents who adopt bed preparation jointly with one of the other techniques is only 34% in the endline survey. Nor do farmers drastically modify traditional cultivation practices (e.g. intercropping rice with root crops at close distances and planting a large number of crops per parcel) that help maintain subsistence and optimize overall production rather than single

crop yields — only 50% of parents use beds while deviating from these traditional practices. These limitations in full-scale adoption, together with a lack of market access, substantially lower the total benefits that farmers can tap from these yield-improving techniques.

While technical training (e.g. management of weeds and pests under new ways of planting) is involved, in this context extension serves mainly to encourage agricultural intensification by demonstrating the possible extent of yield gains. In particular, school farms bear the risks that farmers cannot afford when a bundle of promoted techniques are applied together. By the same token, small-scale projects undertaken by students, who have less obligations to feed the family, can help farmers learn about the returns to a systemic shift to promoted techniques under familiar growing conditions.

The actual yields could, of course, differ from those in agronomic trials.[12] I nonetheless anticipate a net benefit from adoption for the average targeted farmer. In the appendix to Section 5, assuming only half of the yield gains from trials — under the traditional cultivation environment — and accounting for increased labor costs, I estimate a $65 increase in net production value from using beds on a quarter of the cultivated area in one season — roughly 13% of a typical farmer's annual income. This measure omits changes at the intensive margin — the joint adoption of bed preparation with other techniques and under deviations from subsistence practices. Given the lack of trial estimates on the magnitude of yield gains under joint adoption, I maintain an agnostic view of its net effects on welfare but track changes in Section 4.2.1, noting its associations with increases in farm sales and agricultural intensification.

3 Experimental Design

3.1 Sampling and randomization

Sampling frame. The study takes place in five major food-producing counties of Liberia: Bong, Lofa, Nimba, Gbarpolu, and Margibi. Of the 1,794 junior high schools in these counties, 304 met two eligibility criteria verified by field inspection: land and a water source for a school farm.[13] Eligible schools located within two kilometers of one another were grouped into 206 geographical clusters, so that schools in different clusters are at least two kilometers apart, which limits spillovers between SBAE and control schools. From this frame I randomly drew 197 schools (155 clusters) as the study sample; the program's budget allowed 100 SBAE schools.

First stage: SBAE. Randomization was at the cluster level. Clusters were matched into pairs on extension-officer region and average school enrollment in 2015/16 (one group is a triplet), and within each pair one cluster was assigned to SBAE and the other to control. This yielded 100 SBAE schools in 78 clusters and 97 control schools in 77 clusters. All schools assigned to SBAE accepted the program. Figure 1 summarizes the sampling and the two layers of randomization, Figure 2 maps the schools, and Appendix Figure A1 traces participant flow from allocation to analysis.

Second stage: parental engagement. Within SBAE schools, the two parental engagement treatments were randomized independently at the school level, each stratified by district.

- *Program promotional session.* Among the 83 SBAE schools that are not faith-based, 50 were assigned to host promotional video sessions for every sampled parent during the first rainy season of 2021. The

[12] Suri (2011) provides evidence in Kenya that an agricultural technology that offers large average returns in a population can nonetheless have much lower returns for non-adopters. Focusing on common farming techniques promoted in agricultural intensification, Udry (2024) discusses recent evidence on the drastic difference in yields between agronomic trials and general population in Northern Ghana.

[13] Schools were first screened for a functional parent-teacher association and a water source using the national Water, Sanitation, and Hygiene (WASH) dataset (447 schools), then inspected in the field (304 schools). Online Appendix Table OA1 compares sampled schools with other schools in the five counties; eligible schools are larger and have lower student-teacher ratios than the county average.

17 faith-based schools were excluded because their parents live far from the school and could not be reached at reasonable cost.[14]

- *Annual farmer field day (FFD).* Among the 85 SBAE schools that had established a school farm by August 2021, 44 were assigned to host a field day in September 2021 and September 2022.

Because eligibility for the FFD depended on timely farm establishment, which was itself lower in session schools (80 percent) than in other SBAE schools (94 percent), the four engagement cells are unbalanced (20, 30, 24, and 26 schools; Online Appendix Figure OA2), and 15 SBAE schools with late farms were never eligible for the FFD. The estimating equations in Section 4.1 therefore include an indicator for late farm establishment, so that FFD effects are identified among schools that were eligible for it and the "SBAE alone" comparison group is not negatively selected on implementation.

Pre-registration. The trial is registered in the AEA RCT Registry (AEARCTR-0006671). The pre-analysis plan, filed before the midline survey, specifies the primary outcomes in Section 3.4, the use of Lee bounds for differential attrition, and the "short" factorial specification in Section 4.1. [PLACEHOLDER: registration date and any deviations from the plan.] Online Appendix Table OA17 reports minimum detectable effects for the main contrasts and for the interaction between the two engagement treatments, which the design is not powered to detect.

3.2 Treatments and implementation

SBAE. SBAE schools receive bi-annual training of two teachers per school on the integrative SBAE model and the promoted techniques, and monthly visits from extension officers (Section 2.3). Teachers and officers then deliver the three components: agriculture and science instruction (lessons during extension visits and experiential instruction in class), extracurricular activities (weekly farm work, bi-weekly club meetings, annual national events), and technology diffusion through the school demonstration farm and student home projects.

Implementation was far from perfect, which matters for interpreting intent-to-treat effects. By the end of the first rainy season, 70 of the 100 SBAE schools had started a school farm; by August 2021, 85 had. The common reasons for delay were teacher turnover, teachers' non-compliance, and damage to farms from grazing animals and theft. [PLACEHOLDER: share of SBAE schools with a functioning farm at midline and endline, by engagement cell; share of sampled students who had joined the program by endline is 81 percent in SBAE schools without engagement (Section 6.1).]

Program promotional session. The session is inspirational rather than technical. About six months into the program, during the first student follow-up survey, enumerators invited one parent per sampled student to a 20- to 30-minute session held outside the school. In groups of four to five, parents watched a video that explains how SBAE integrates experiential science instruction, school farms, extracurricular clubs, and student home projects, and answered survey questions before and after. Half of the parents, randomly selected, watched a version that adds testimonials from past participants, town chiefs, extension officers, teachers, and parents ("the cassava I got was big, big"; "I paid my own school fees"). The video mentions no specific farming technique, so any effect on household farming practices must operate through parents' interactions with students and the program. Both versions raise parents' beliefs about students' farm management abilities immediately after viewing, and these beliefs remain elevated at midline and endline; this paper pools the two versions and the companion paper (JMP2, Lee, 2025) studies them separately. Online Appendix C describes the videos and the immediate belief responses.

Annual farmer field day. The field day is a training intervention focused on one program component: the school farm as a demonstration plot. Each September of the first two program years, extension officers

[14] Faith-based schools cluster geographically, and by chance SBAE received more multi-school clusters than control, so the share of faith-based schools is higher in SBAE (17 of 100) than in control schools (10 of 97). More schools were assigned to the session arm than to its complement because the companion paper (JMP2, Lee, 2025) required a larger sample of parents who watched the video.

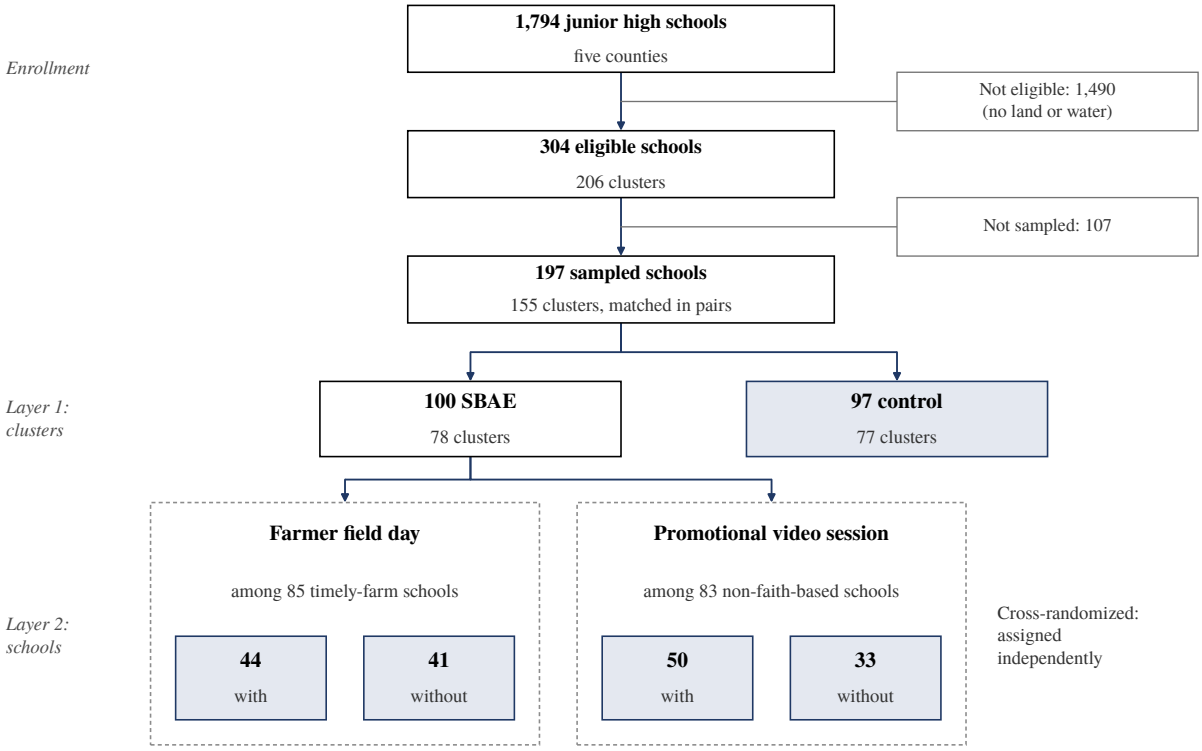


FIGURE 1. EXPERIMENTAL DESIGN OF THE STUDY

Notes: This figure summarizes sampling and the two layers of randomization. Of the 1,794 junior high schools in the five study counties, 304 met the eligibility criteria (land and a water source for a school farm) and were grouped into 206 geographical clusters; 197 schools in 155 clusters were drawn at random and matched into pairs on extension-officer region and enrollment (one triplet). In the first layer, within each pair, one cluster was assigned to SBAE (100 schools, 78 clusters) and the other to control (97 schools, 77 clusters). In the second layer, the two parental engagement treatments were assigned independently at the school level, stratified by district: an annual farmer field day among the 85 SBAE schools with a school farm by August 2021, and a program promotional video session among the 83 SBAE schools that are not faith-based. Shaded boxes are the treatment cells used in estimation. Online Appendix Figure OA2 shows the full assignment tree by school type and farm status, and Appendix Figure A1 reports participant flow.

harvested part of the root crops to show the gains from bed preparation and explained the promoted techniques. Town chiefs were asked to select 25 to 30 farming-group leaders per school to attend alongside parents; each event lasted about six hours. Attendance was not recorded, but Section 6 shows that the field day raises the share of parents who have observed a harvest on the school farm and improves their beliefs about attainable cassava yields.

The two treatments thus target distinct gaps. The video addresses how parents understand the program and the value of students' learning under it; the field day transfers technical knowledge that parents can apply while keeping students as farm assistants. The companion paper shows that the field day alone leads parents to use students as assistants on subsistence parcels rather than as managers of sales-oriented parcels.

3.3 Timeline and data

The program ran from January 2021, the delayed start of the 2020/21 school year, to December 2023, covering three school years and three rainy seasons (Figure 3). Each year a survey round took place in the middle of the rainy season, which coincides roughly with the end of the school year and with land preparation, when adoption of soil management techniques can be observed. The first follow-up survey (2021) captures SBAE's

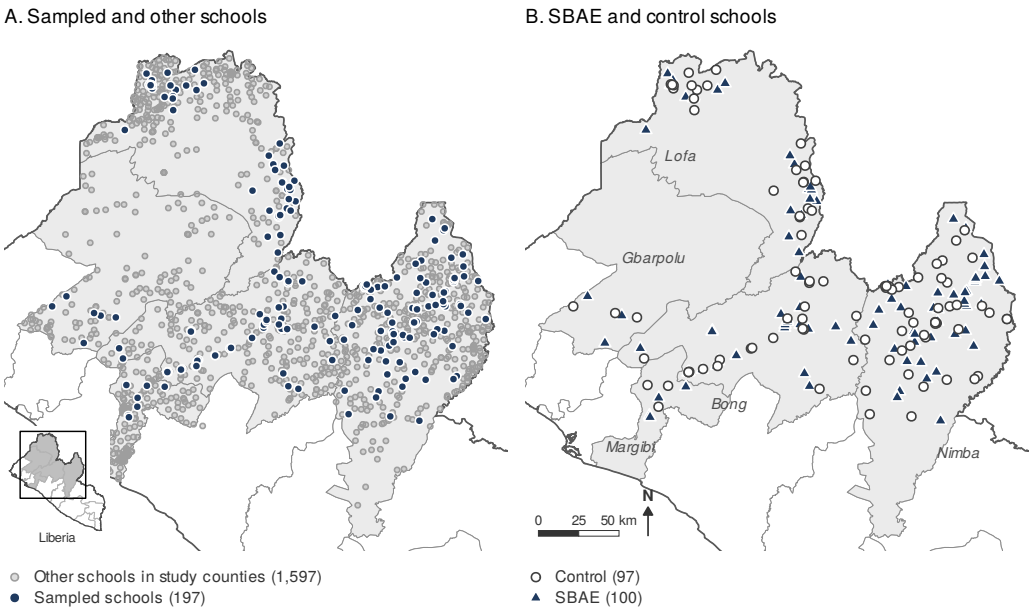


FIGURE 2. GEOGRAPHICAL DISTRIBUTION OF SAMPLED SCHOOLS

Notes: Panel A shows the 197 sampled junior high schools and the 1,597 other schools in the five study counties (Bong, Gbarpolu, Lofa, Margibi, and Nimba), based on Ministry of Education school records. Panel B shows the 100 SBAE and 97 control schools. Study counties are shaded; the inset locates the study area within Liberia. County boundaries are from OCHA (2019).

early effects on students before the engagement treatments took effect; the midline (2022) and endline (2023) surveys measure the primary outcomes. Table 1 summarizes the samples by round.

Students. The program recruits students in upper primary and junior high grades. The sample is drawn from grades 4, 5, 7, and 8, excluding grades 6 and 9, whose students leave the school at the end of the year. Within each school, enumerators screened students in these grades during 2020/21 and selected about 19 students who farmed outside school and whose parents farmed, prioritizing students with prior farming and cooperative-group (kuu) experience, who a pilot showed were more likely to participate in SBAE.[15] The resulting 3,757 students, one per household, are 30 percent of students in the targeted grades and closely resemble all eligible students: 99.4 versus 95.0 percent had farmed before, and 82.1 versus 77.1 percent had joined a kuu (Online Appendix Table OA1, Panel B).

Parents. For each sampled student, one parent or guardian (henceforth, parent) who planted root crops or vegetables in 2021 was invited; where both parents qualified, the invitee was chosen at random to balance gender. Because of budget constraints, parents were interviewed only at endline, in a random 45 percent of households. In 95 percent of cases the invited parent participated; otherwise another household member did.

Teachers. To measure school-level changes in instruction, all science and agriculture teachers in each school were interviewed at midline (452) and endline (472) as repeated cross-sections; teacher turnover in rural Liberia is too high to follow individuals.

Survey content. Student surveys cover schooling, time use, livelihoods, aspirations, beliefs, and a test of knowledge about the promoted techniques; parent surveys cover engagement with the school, beliefs, and

[15] Enumerators worked down a priority list (farmed and joined a kuu before 2021; joined a kuu; farmed; other eligible students) until reaching the school's target, which was raised in larger schools and lowered in smaller ones. Within priority tiers, selection balanced section (upper primary or junior high) and gender.

TABLE 1—STUDY SAMPLES BY SURVEY ROUND

	Baseline 2020/21	First follow-up 2021	Midline 2022	Endline 2023
Panel A. Schools				
Control schools (clusters)	97 (77)	97 (77)	97 (77)	97 (77)
SBAE schools (clusters)	100 (78)	100 (78)	100 (78)	100 (78)
with promotional session	–	50	50	50
with farmer field day	–	44	44	44
with both	–	20	20	20
Panel B. Students				
Sampled	3,757	–	–	–
Targeted for interview	–	–	3,757	3,757
Interviewed	–	2,520	2,425	1,488
of which, farm module	–	–	1,388	–
Panel C. Parents				
Targeted for interview	–	–	–	1,691
Interviewed	–	–	–	1,543
of which, farm module	–	–	–	1,363
Panel D. Teachers				
Interviewed (repeated cross-section)	–	–	452	472

Notes: The first follow-up targeted all sampled students in the 50 session schools and a random half of sampled students in the other 147 schools. The endline parent survey targeted a random 45 percent of sampled households, stratified by school. Farm modules were administered to a random subset of interviewed subjects. Parent counts in the "targeted" row are approximate; [PLACEHOLDER: replace with exact counts from the tracking data.]

support for students' education. A parcel-level farm module, administered to a random subset at midline (students) and endline (students and parents), measures technology adoption on one "student parcel" and, at endline, one "parent parcel," chosen at random with priority to parcels the respondent looks after and to parcels planted with root crops or vegetables.

3.4 Primary outcomes

Online Appendix D defines all variables. The primary outcomes, pre-specified, are:

- *Agricultural technology adoption*, on students' parcels (midline and endline) and parents' parcels (endline): whether bed preparation is used for root crops or vegetables, and the number of the four promoted techniques (Section 2.4) in use.
- *School attendance* (midline and endline): whether the student has dropped out, according to school records and household tracking; days absent in the last five school days; and imputed total days of attendance over 2021/22 and 2022/23.[16]
- *Studying effort*: hours spent studying in the week before the survey (midline and endline).
- *Livelihoods* (endline, retrospective): annual farm and non-farm income and annual savings, log-transformed and winsorized at 2.5 percent in each tail; meals missed in the past week; and non-food spending in the past six months on inputs, schooling, and consumption goods.
- *Aspirations*: whether an agricultural career is among the student's top three unprompted job choices (midline), and whether the student saves for university (endline).

[16] Imputed from enrollment status, the absence rate, and 200 school days per year. Dropouts are assigned zero days; students who switched schools are assigned the control-school mean, which is conservative for the treatment effect because switching is more common in the treatment arms with lower dropout.

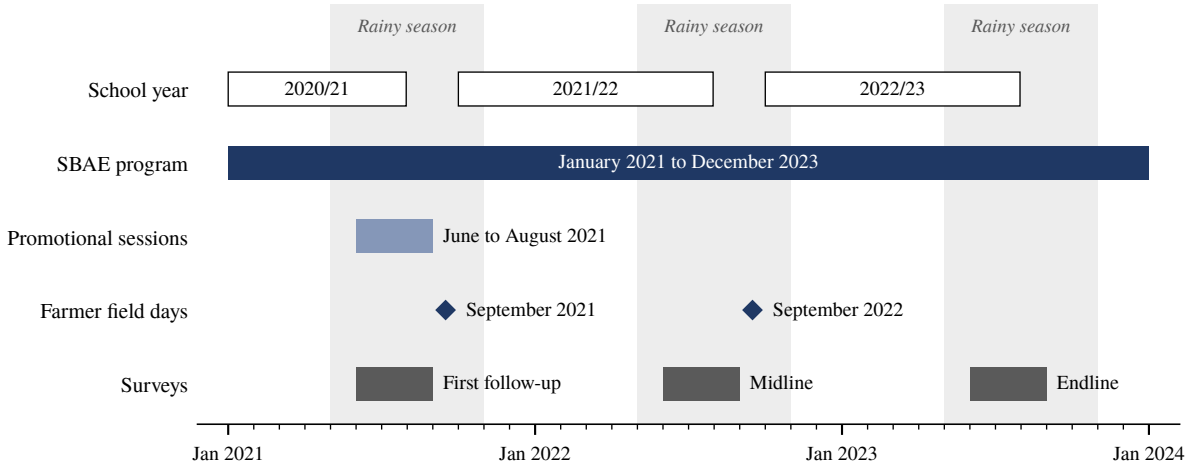


FIGURE 3. TIMELINE OF SBAE PROGRAM AND DATA COLLECTION ACTIVITIES

Notes: Shaded bands mark the rainy seasons (April–October). The 2020/21 school year began in January 2021, three months late, because of COVID-19 school closures. Promotional sessions were held for sampled parents in 50 SBAE schools during the first follow-up survey; farmer field days were held in 44 SBAE schools in September 2021 and September 2022. The first follow-up survey covered students; the midline and endline surveys covered teachers, students, and parents.

Secondary outcomes used in Section 6 to trace mechanisms are beliefs about attainable cassava yields, knowledge of the promoted techniques, teachers' pedagogy, parents' visits to school farms, students' management of farms, students' demonstration of techniques to parents, parents' support for students' education, and joint adoption of techniques.

3.5 Balance and attrition

Balance. Online Appendix Table OA2 regresses baseline characteristics on the treatment indicators of equation (1b), with randomization-pair fixed effects and standard errors clustered by geographical cluster. School characteristics in 2020/21 (enrollment, teachers, classrooms, water source, meals, management, ownership) and student characteristics at sampling (age, grade, gender, farming experience) are balanced; no joint F-test rejects at the 10 percent level. Parents were first interviewed at endline, so the parent panel reports time-invariant or pre-determined characteristics (age, gender, relationship to the student, household headship, literacy, and schooling) in the endline sample; these are also balanced.

Attrition. Student attrition is high: 35 percent at midline and 48 percent at endline (Appendix Table A1), driven by dropouts (18 and 24 percent) and transfers to other schools (16 and 21 percent), the latter mostly from mechanical progression to junior or senior high. Neither SBAE nor the field day changes attrition, but the promotional session lowers it by 7.3 points at midline and 9.2 points at endline, entirely through fewer dropouts and transfers. Parent attrition at endline is 12 percent; SBAE raises it by 4.2 points and the field day lowers it by 4.0 points. Attrition is selective on age and grade for students and on age and guardian status for parents, but Wald tests show that this selection does not differ across arms (Online Appendix Table OA3). I therefore report Lee (2009) bounds alongside every main estimate, show that effects on key outcomes are consistent across the three rounds (Sections 6.1.2 and 6.3), and show in the Online Appendix that effects are similar for upper- and lower-grade students.

4 Experimental results

4.1 Empirical specification

I analyze SBAE's impact on the primary outcomes: agricultural technology adoption, school attendance, students' livelihoods, and students' aspirations. I estimate intent-to-treat (ITT) effects from

$$Y_{hscp} = \alpha_1 + \beta_1 \,\text{SBAE}_p + \gamma_p + \varepsilon_{hscp}, \tag{1a}$$

$$Y_{hscp} = \alpha_2 + \beta_1 \,\text{SBAE}_p + \beta_2 \,\text{SBAE}_p \times \text{FFD}_s + \beta_3 \,\text{SBAE}_p \times \text{Session}_s + \lambda \,\text{Faith-based}_s + \mu \,\text{SBAE}_p \times \text{Late farm}_s + \gamma_p + \varepsilon_{hscp}, \tag{1b}$$

where Y_{hscp} is the outcome for household h in school s, geographical cluster c, and randomization pair p; SBAE_p indicates that pair p's treated cluster contains school s; FFD_s and Session_s indicate assignment to the farmer field day and the promotional session; Faith-based_s indicates a faith-based school, which was ineligible for the session; Late farm_s indicates an SBAE school that had not established its farm by August 2021 and was therefore ineligible for the field day; and γ_p are randomization-pair fixed effects. The two eligibility indicators absorb the non-random exclusions described in Section 3.1, so that β_1 is the effect of SBAE alone in schools that were eligible for both engagement treatments but received neither, and β_2 and β_3 are identified from random assignment among eligible schools. Standard errors are clustered at the geographical-cluster level, the highest level at which assignment took place.

(1a) estimates the overall SBAE effect, disregarding differences in parental treatment conditions among SBAE schools. (1b) estimates the distinct effect of each treatment combination. I present results from both regressions for the following reason. Regression (1a) contains the overall SBAE treatment effect, β_1, which is a weighted average of the effects in SBAE schools assigned to different parental engagement conditions as captured by β's in (1b). On the one hand, if we treat SBAE schools asssigned to different conditions to be the same, then β_1 in (1a) provides a more precise estimate of SBAE's effects. On the other hand, SBAE schools assigned to different conditions can have varying effects — in this case we will want to see which combination of SBAE and parental engagement is sufficient to improve outcomes, thus the need of (1b).

To be specific, we will discuss the following estimates of interest: (i) overall SBAE effect (β_1 in (1a)); (ii) pure SBAE effect (β_1 in (1b)); (iii) "SBAE + FFD" effect (β_1 + β_2 in (1b)); (iv) "SBAE + Session" effect (β_1 + β_3 in (1b)); and (v) "SBAE + Session + FFD" effect (β_1 + β_2 + β_3 in (1b)).

To maximize statistical power, (1b) uses what Muralidharan et al. (2025) refer to as the "short model" in factorial design. β_2 and β_3 give the marginal effects of the promotional session and the annual FFD in a sample in which (roughly) half the households are offered the other treatment. We are underpowered to detect interaction effects. In the Online Appendix, I show that the main point estimates are similar when using the "long model" that controls for an interaction term between the two parental engagement treatments. Following Kaur et al. (2025), In the Online Appendix I also show that the effects of the promotional session (FFD) are qualitatively similar in the sub-samples with and without FFD (promotional session).

I present the results in two steps. Section 4.2 contrasts SBAE with and without parental engagement (β_1 versus β_1 + β_2 + β_3), outcome by outcome. Section 4.3 examines the marginal effects of the program promotional session and the annual farmer field day (β_3 and β_2). Section 6 returns to the standalone effect of SBAE (β_1 in (1b)) and to the mechanisms behind the pivotal role of parental engagement.

4.2 SBAE with and without parental engagement

Figures 4 to 8 contrast SBAE with and without parental engagement. For each primary outcome, they plot two estimates from (1b): the pure SBAE effect ("without engagement", β_1) and the effect of SBAE augmented with both the program promotional session and the annual farmer field day ("with engagement", β_1 +β_2 +β_3), each with its 95% confidence interval and Lee bounds. Full regression estimates—including the overall

SBAE effect from (1a) and the "SBAE + FFD" and "SBAE + Session" combinations—are reported in Online Appendix Tables OA4 to OA10.

4.2.1 Agricultural technology adoption and agricultural intensification

We begin by presenting the results on households' technology adoption in Figure 4, with full estimates in Online Appendix Table OA4.

Figure 4 shows that parental engagement transforms SBAE's effects on technology adoption. Without parental engagement, the effects on the student parcel are small and statistically insignificant at both midline and endline, and on the parent parcel SBAE raises only the number of techniques used. With both forms of parental engagement, SBAE raises students' use of bed preparation by 15.0 percentage points (33% from 45% in control schools) and the number of techniques used by 0.39 (27% from 1.43) in the midline, and raises them by 18.2 percentage points (63% from 29%) and 0.39 (53% from 0.73) in the endline. On the parent parcel, it raises the use of bed preparation by 10.9 percentage points (15% from 74% in control schools) and the number of techniques used by 0.36 (26% from 1.38 in control schools). All these strong effects are preserved when considering Lee lower bounds. Alternatively, the augmented SBAE program raises the number of techniques used by parents by 0.38 standard deviations — a sizable effect considering that parents are not directly trained in the program and that households do not receive other forms of interventions (e.g. provision of seeds or credit) than provision of knowledge and skills. By comparison, a large literature on sub-Saharan Africa (Suri and Udry, 2022) suggests that many village-based agricultural extension fails to create statistically significant impacts on farmers who are not directly trained, unless social network targeting (Beaman et al.,

2021) or incentives for trained farmers to communicate with others (BenYishay and Mobarak, 2019) are built into such extension programs. Figure 4 thus provides strong proof-of-concept that school-based agricultural extension can be a highly effective method in increasing technology diffusion.

While Figure 4 focuses on the primary measures of adoption, Online Appendix Table OA5 indicates the augmented SBAE program is deepening agricultural intensification in a comprehensive way. I consider two secondary measures of agricultural intensification: first, whether the main yield-improving technique, bed preparation, is applied jointly with at least one of the three other soil management techniques, noting the potential of such pairing to increase yields; second, whether bed preparation is applied jointly with at least one deviation from three traditional cultivation practices (fallowing in previous rainy season; cultivation of roots or vegetables on rice fields; and intercropping with rice), noting the potential of such pairing to increase sales.

Whereas Figure 4 suggests that the SBAE program alone is entirely ineffective in increasing the use of bed preparation, Online Appendix Table OA5 provides a slightly different picture. Columns (6) and (10) indicate that the SBAE program, on its own, increases both students' and parents' joint use of bed preparation with other promoted techniques — by 8.8 percentage points for students (from 17%) and by 7.6 percentage points for parents (from 34%). By contrast, the SBAE program, on its own, has close to zero and statistically insignificant effects on whether students and parents use bed preparation along with deviations from traditional cultivation practices (columns (8) and (12)). While the program promotional session intensifies students' and parents' use of bed preparation along with other techniques (columns (2) and (10); for students, only at midline), its main effect is to dramatically increase the use of bed preparation in settings deviating from traditional cultivation practices — by 12.1 percentage points for students (from 25%; column (8)) and 9.8 percentage points for parents (from 52%; column (12)). As in the case of primary measures of technology adoption, the annual farmer field day does not affect these secondary measures of technology adoption at endline. In summary, Online Appendix Table OA5 suggests that besides pivotally increasing the use of promoted techniques, the program promotional session also dramatically increases the use of yield-improving techniques in settings that are more amenable to sales.

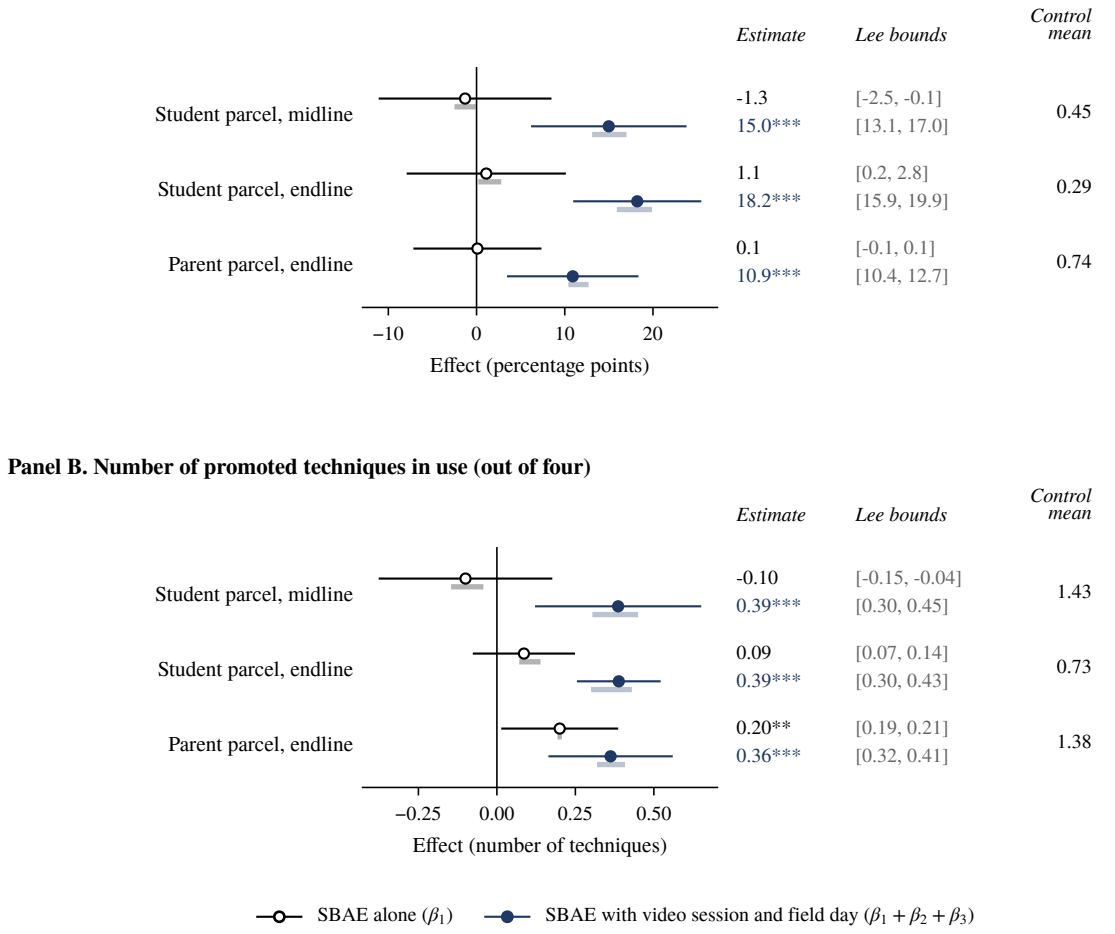


FIGURE 4. MAIN RESULTS: AGRICULTURAL TECHNOLOGY ADOPTION

Notes: This figure plots intent-to-treat effects estimated from (1b). "Without engagement" is the pure SBAE effect (β_1); "with engagement" is the effect of SBAE augmented with both the program promotional session and the annual farmer field day ($\beta_1 + \beta_2 + \beta_3$). Markers are point estimates and horizontal lines are 95% confidence intervals, based on standard errors clustered at the geographical-cluster level. Gray bands beneath the confidence intervals mark Lee (2009) lower and upper bounds. All specifications include pair-of-clusters fixed effects and an indicator for faith-based schools. Significance: *** p<0.01, ** p<0.05, * p<0.10. Outcomes are measured on the designated student parcel (midline and endline) and parent parcel (endline); "number of techniques" counts the four promoted soil management techniques. Full estimates are in Online Appendix Table OA4.

4.2.2 School attendance and time use

Figure 5 presents results on students' school attendance, with full estimates in Online Appendix Table OA6.

Figure 5 contrasts the effects of SBAE with and without parental engagement on attendance. These estimates provide three messages. First, without parental engagement, SBAE has no effect on dropouts, and the Lee lower bounds confirm that the SBAE program on its own increases student absenteeism. Second, including both types of parental engagement in the SBAE program reduces dropouts by 3.8 percentage points in the midline and 3.9 percentage points in the endline (from 22% and 33% in control schools). These point estimates are imprecise; the Lee bounds range from 3.2 to 6.7 percentage points in the midline and from 2.7 to 6.8 percentage points in the endline.[17] Third, including both types of parental engagement is sufficient to

[17]There is a drop in the magnitude and precision of point estimates when moving from "SBAE + Session" to "SBAE + Session + FFD".

overturn the pure SBAE effect on absenteeism — reducing absenteeism by 18% in the midline (a drop of 0.12 from 0.69 out of 5 days) and by 27.8% in the endline (a drop of 0.18 from 0.65 out of 5 days); the endline reduction is statistically significant and robust to Lee bounds.

The last row of Figure 5 uses a single imputed measure of school attendance to account for the effects on both students' school enrollment and absenteeism. This measure counts the total number of school days students attend in the school year 2021/22 (year of the midline) to 2022/23 (year of the endline) with three imputation rules: first, days of absence are subtracted from a school year of 200 school days, assuming that the rate of absenteeism is constant throughout the year; second, students who have dropped out by the midline and endline surveys are counted as having zero school days in the corresponding school year; third, students who have switched schools are counted as having the same days of absence as the average among control school students who respond to surveys. Without parental engagement, SBAE has a small and statistically insignificant effect on overall school attendance (a drop of 3.6 days per student). The combined estimate of the SBAE program with both types of parental engagement indicates an increase in overall school attendance by 17.6 days per student (8% from 229.1 days in control schools). I will use this number in cost-benefit and cost-effectiveness analyses in Section 5.

Figure 6 presents results on students' time use (full estimates in Online Appendix Table OA7), with treatment effects on studying hours that mirror those on absenteeism.

Figure 6 provides two messages. First, the SBAE program on its own reduces students' studying hours; in the endline, even the Lee upper bound is significantly negative. Second, including both types of parental engagement is sufficient to overturn the pure SBAE effect — increasing students' studying hours by 7% in the midline (statistically insignificant) and by 21.7% in the endline (an increase of 1.17 hours from 5.39 hours per week).

To complete the description of students' time use, the last two rows of Figure 6 show students' weekly hours spent on farming (inclusive of hours spent on school farms). Neither SBAE alone nor SBAE with both types of parental engagement increases students' farming hours significantly relative to control schools — rebutting concerns that SBAE might increase child labor.

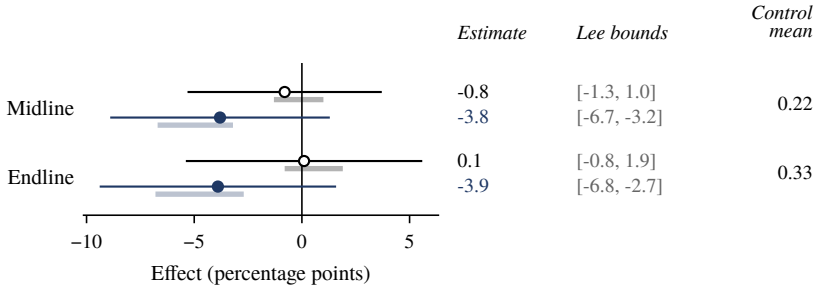
4.2.3 Students' livelihoods and aspirations

Figure 7 presents results on students' farm sales, non-farm income, total income, and savings in 2022, retrospectively reported in the endline survey, with full estimates in Online Appendix Table OA8. For each outcome, I estimate effects at the extensive margin (whether students have positive values), the intensive margin (log values among students with positive outcomes), and overall percentage changes using Poisson regression. All monetary values are expressed in 2022 Liberian dollars and winsorized at the 2.5 and 97.5 percentiles.

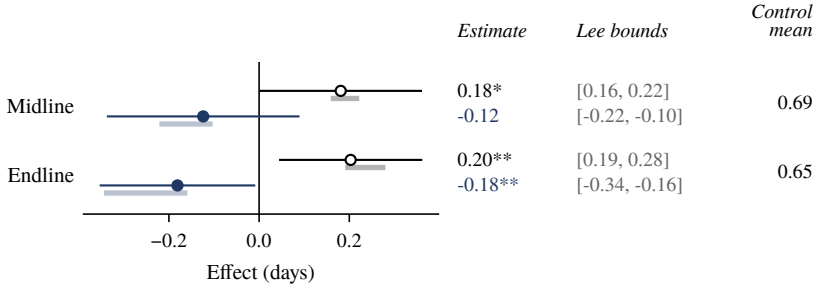
Figure 7 shows that, without parental engagement, SBAE has no significant effect on students' farm sales, reduces the fraction of students with non-farm income by 6.1 percentage points, and has no significant effect on total income or savings. With both types of parental engagement, SBAE increases the fraction of students with farm sales by 9.6 percentage points (14% from 70% in control schools), with a statistically insignificant 14% increase in farm sales in the Poisson specification (from 3,407 Liberian dollars). Its effects on non-farm income and total income are small and statistically insignificant. Most notably, it increases the fraction of students with positive savings by 10.7 percentage points (19% from 55% in control schools) and total savings by 21% in the Poisson specification (significant at the 10% level; from a baseline of approximately US$20 annually).

This is caused by a negative but imprecise effect of the annual farmer field day on enrollment. In view of this, I take the effect of "SBAE + Session + FFD" (3.8 percentage points in the midline and 3.9 percentage points in the endline) to be the lower bound and the effect of "SBAE + Session" (4.5 percentage points in the midline and 5 percentage points in the endline) to be the upper bound of the true effect of the augmented SBAE program.

Panel A. Dropped out of school (percentage points)



Panel B. Days absent in the last five school days



Panel C. Total days of school attendance, 2021/22 to 2022/23 (imputed)

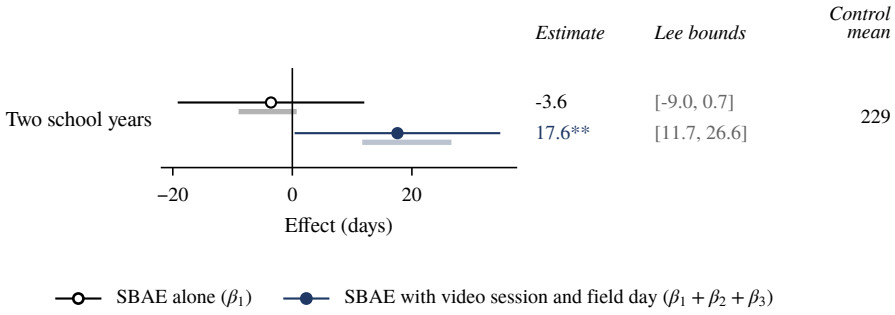


FIGURE 5. MAIN RESULTS: SCHOOL ATTENDANCE

Notes: This figure plots intent-to-treat effects estimated from (1b). "Without engagement" is the pure SBAE effect (β_1); "with engagement" is the effect of SBAE augmented with both the program promotional session and the annual farmer field day ($\beta_1 + \beta_2 + \beta_3$). Markers are point estimates and horizontal lines are 95% confidence intervals, based on standard errors clustered at the geographical-cluster level. Gray bands beneath the confidence intervals mark Lee (2009) lower and upper bounds. All specifications include pair-of-clusters fixed effects and an indicator for faith-based schools. Significance: *** p<0.01, ** p<0.05, * p<0.10. "Dropped out" is one minus the enrollment indicator in Online Appendix Table OA6, so its estimates and Lee bounds have the opposite sign to those in the table. Total days of attendance over 2021/22–2022/23 is imputed as described in Section 3.4.1. Full estimates are in Online Appendix Table OA6.

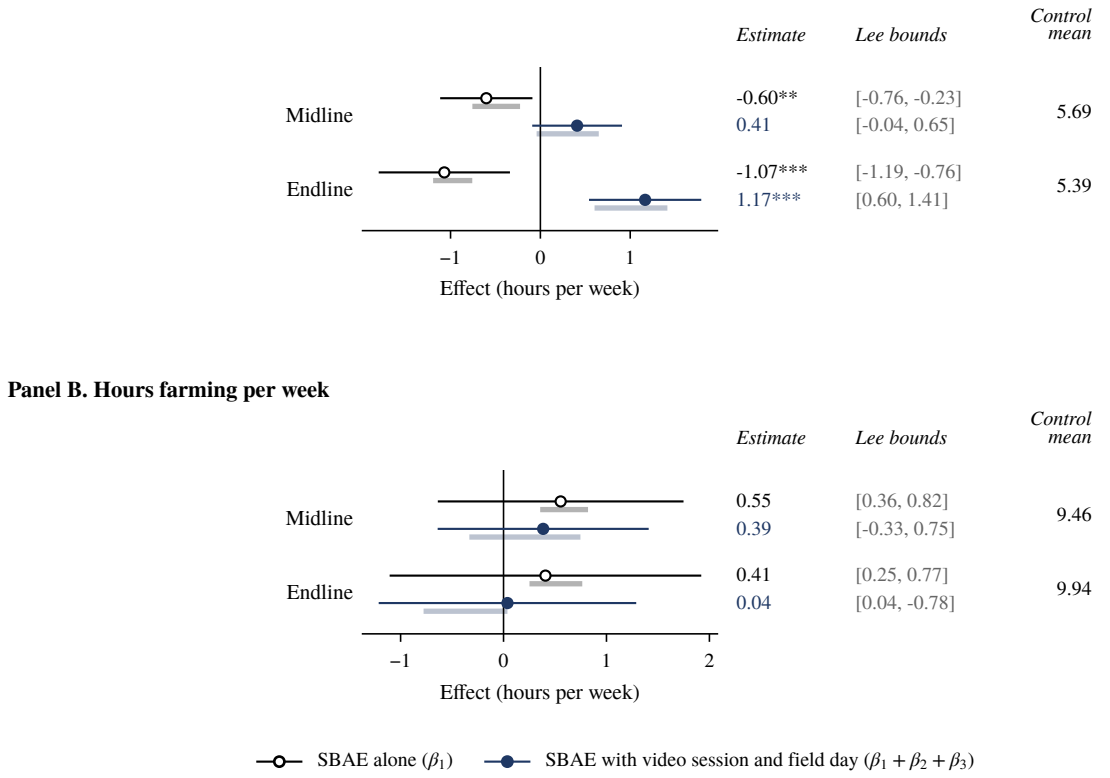


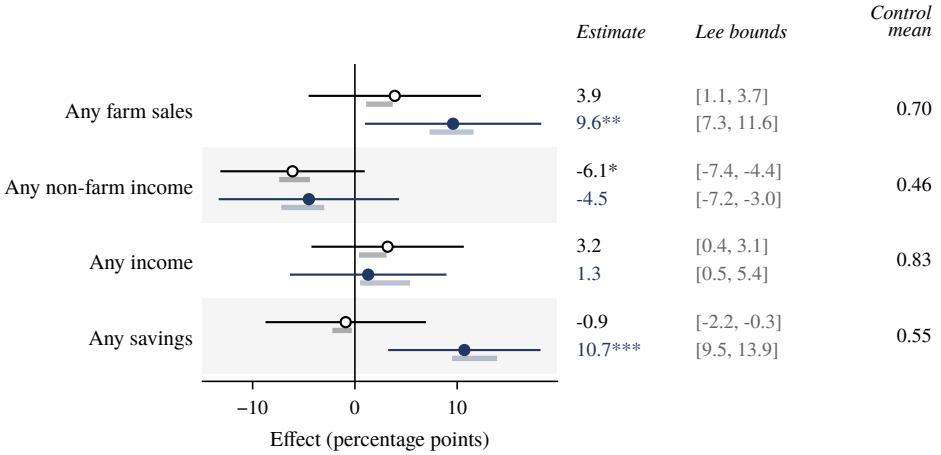
FIGURE 6. MAIN RESULTS: STUDENTS' TIME USE

Notes: This figure plots intent-to-treat effects estimated from (1b). "Without engagement" is the pure SBAE effect (β_1); "with engagement" is the effect of SBAE augmented with both the program promotional session and the annual farmer field day ($\beta_1 + \beta_2 + \beta_3$). Markers are point estimates and horizontal lines are 95% confidence intervals, based on standard errors clustered at the geographical-cluster level. Gray bands beneath the confidence intervals mark Lee (2009) lower and upper bounds. All specifications include pair-of-clusters fixed effects and an indicator for faith-based schools. Significance: *** p<0.01, ** p<0.05, * p<0.10. Hours are for the week preceding each survey; farming hours include hours on school farms. Full estimates are in Online Appendix Table OA7.

While students' increases in farm sales tend to be offset by decreases in non-farm income, our measurement of total income likely understates actual gains from the program, since many students work as assistants to household members and do not directly sell farm produce. Savings provide a better measure of students' economic gains in this context, as they reflect transfers from household members in addition to direct earnings. The combined parental engagement interventions significantly increase savings beyond the SBAE-only effect (p=0.04 for the joint test). Online Appendix Table OA9 confirms that this increase in savings stems from higher income rather than reduced consumption—the SBAE program has close to zero and statistically insignificant effects on students' overall spending (Poisson specification) and meal consumption, with or without parental engagement.

Turning to the last category of primary outcomes, Figure 8 shows results on students' aspirations in the midline and endline surveys, with full estimates in Online Appendix Table OA10. Parental engagement plays a pivotal role in increasing students' study of agriculture as an elective subject: SBAE alone has no significant effect, whereas SBAE with both types of parental engagement increases it by 18.4 percentage points (from 50% in control schools). SBAE alone increases students' naming of agricultural careers by 7.8 percentage points (from 43% in control schools), while including both types of parental engagement intensifies this effect

Panel A. Any income or savings in the past year (percentage points)



Panel B. Amounts in Liberian dollars (Poisson coefficients)

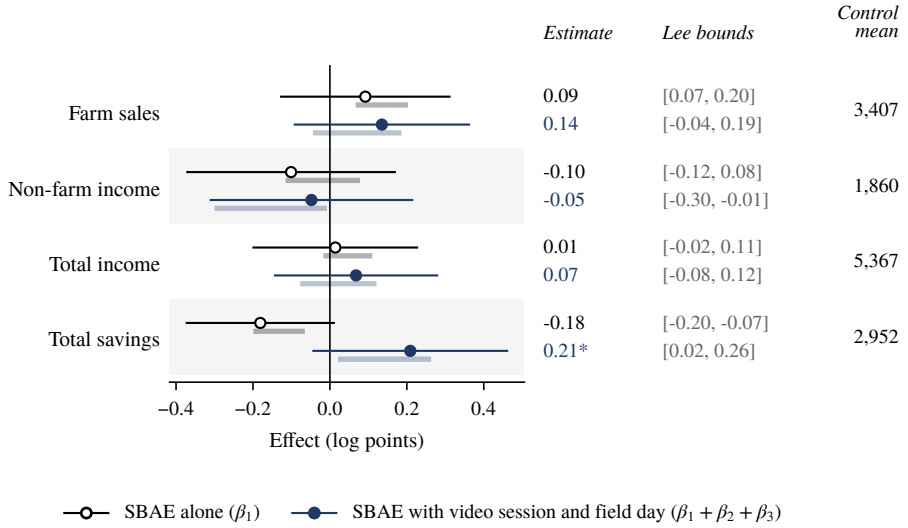


FIGURE 7. MAIN RESULTS: STUDENTS' INCOME AND SAVINGS

Notes: This figure plots intent-to-treat effects estimated from (1b). "Without engagement" is the pure SBAE effect (β_1); "with engagement" is the effect of SBAE augmented with both the program promotional session and the annual farmer field day ($\beta_1 + \beta_2 + \beta_3$). Markers are point estimates and horizontal lines are 95% confidence intervals, based on standard errors clustered at the geographical-cluster level. Gray bands beneath the confidence intervals mark Lee (2009) lower and upper bounds. All specifications include pair-of-clusters fixed effects and an indicator for faith-based schools. Significance: *** p<0.01, ** p<0.05, * p<0.10. Outcomes refer to the midline year (2022), reported retrospectively at endline, in Liberian dollars (L$) winsorized at the 2.5th and 97.5th percentiles. "Any" outcomes are extensive-margin indicators estimated by OLS; the other outcomes are estimated by Poisson regression, whose coefficients approximate proportional changes and are reported directly in the "vs. control" column. Intensive-margin estimates and full estimates are in Online Appendix Table OA8.

to 11.5 percentage points over control schools. Finally, SBAE alone has no significant effect on students' saving for university, whereas SBAE with both types of parental engagement increases it by 6.4 percentage points (from 17% in control schools) — an effect driven by the annual farmer field day (Section 4.3).

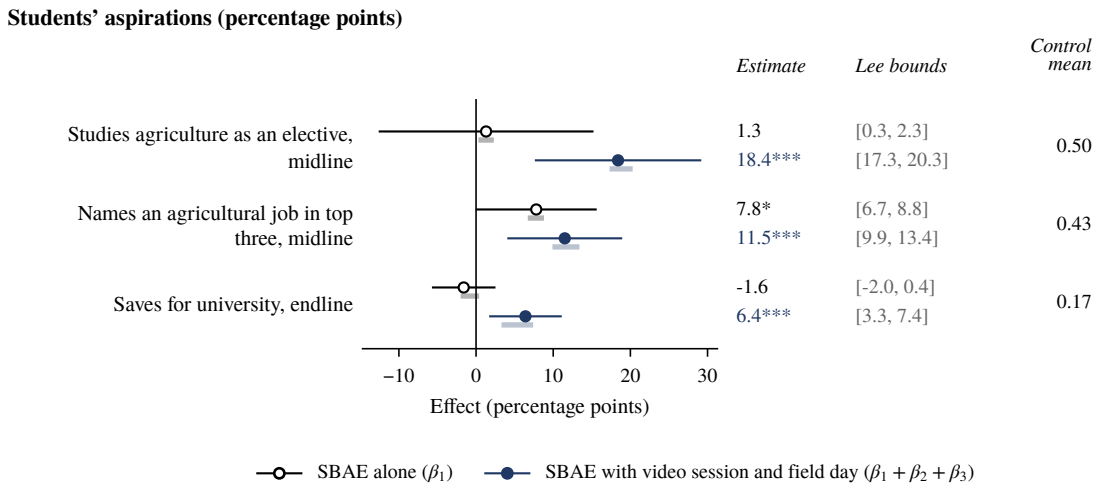


FIGURE 8. MAIN RESULTS: STUDENTS' ASPIRATIONS

Notes: This figure plots intent-to-treat effects estimated from (1b). "Without engagement" is the pure SBAE effect (β_1); "with engagement" is the effect of SBAE augmented with both the program promotional session and the annual farmer field day ($\beta_1 + \beta_2 + \beta_3$). Markers are point estimates and horizontal lines are 95% confidence intervals, based on standard errors clustered at the geographical-cluster level. Gray bands beneath the confidence intervals mark Lee (2009) lower and upper bounds. All specifications include pair-of-clusters fixed effects and an indicator for faith-based schools. Significance: *** p<0.01, ** p<0.05, * p<0.10. Agricultural jobs include farmer, agronomist and extension officer, named without prompting. Full estimates are in Online Appendix Table OA10.

4.2.4 Summary and robustness to multiple hypothesis testing

In summary, Figures 4 to 8 all suggest the pivotal role of parental engagement in SBAE's effectiveness. Whereas SBAE on its own falls short of its goals (Section 6.3), SBAE augmented with both types of parental engagement is sufficient to achieve all desired goals: (i) increasing parents' adoption of promoted techniques by 0.38 standard deviations; (ii) reducing absenteeism by 27.8% and dropouts by an estimated 3.8 to 5 percentage points, although the dropout estimates are imprecise; (iii) increasing students' weekly studying hours by 21.7%; (iv) increasing the fraction of students with savings by 10.7 percentage points and their total savings by 21% without sacrificing their spending and meals consumption; and (v) increasing the fraction of students who find an agricultural career favorable and save for attending university.

To economize on space, Figures 4 to 8 and the corresponding appendix tables have not shown robustness checks with respect to multiple hypothesis testing. Online Appendix Figure OA1 addresses this issue by plotting the naive p-values and adjusted q-values based on the procedure of Anderson (2008), which controls the false discovery rate across multiple tests. To be specific, I conduct two types of adjustments: (i) pooling all 18 outcomes together and adjusting for the four linear combination coefficients tested (SBAE, SBAE + Session, SBAE + FFD, and SBAE + Session + FFD) — thus 72 tests in total, disregarding potential correlations between outcomes, such as between students' and parents' adoption; (ii) pooling only outcomes within pre-specified categories: testing 6 outcomes for 4 coefficients under "technology adoption"; testing 3 outcomes for 4 coefficients under "school attendance and learning"; testing 2 outcomes for 4 coefficients under "income and savings"; testing 4 outcomes for 4 coefficients under "spending and food security"; and

testing 3 outcomes for 4 coefficients under "aspirations".

Online Appendix Figure OA1 presents the robustness of ten main results for the augmented SBAE program with program promotional session and the annual farmer field day. Seven out of ten results have p-values below 0.01 before correction, and seven out of ten have adjusted q-values below 0.05 under both adjustments. For the remaining results (total days of attendance, farm sales, savings), the adjusted q-values range from 0.10 to 0.67. Hence the effects of the augmented program on technology adoption, studying hours, and aspirations are robust to multiple hypothesis testing, whereas its effects on total days of attendance and on students' farm sales and savings should be interpreted with more caution.

4.3 Marginal effects of the promotional video and the farmer field day

Figure 9 separates the contributions of the two forms of parental engagement. For every primary outcome, it plots the marginal effect of the program promotional session (β_3 in (1b)) and of the annual farmer field day (β_2), each scaled by the control-school mean.

The program promotional session, which is experimentally designed to reach every parent, plays an especially important role in parental engagement (Panel A of Figure 9). First, it raises technology adoption on all six measures. On the student parcel, it raises both the use of bed preparation and the number of techniques used in the midline (by 10.8 percentage points from 45% in control schools, and by 0.28, 20% from 1.43) and in the endline (by 10.4 percentage points from 29%, and by 0.18, 24% from 0.73). It is also the only treatment that raises adoption on the parent parcel — increasing the use of bed preparation by 8.8 percentage points (from 74% in control schools) and the number of techniques used by 0.24 (18% from 1.38 in control schools). Second, only the promotional session reduces dropouts, by 5 percentage points by the endline (from 33% in control schools); its effects on absenteeism are negative but statistically insignificant; and it is the only treatment that increases overall school attendance, by 19.2 days per student. Third, it increases students' weekly studying hours by 12.4% in the midline (0.71 hours from 5.69 hours per week) and by 20.7% in the endline (1.12 hours from 5.39 hours per week), without increasing their farming hours. Fourth, only the promotional session increases the fraction of students with farm sales (by 8.4 percentage points, 12% from 70% in control schools), while its estimated effect on students' non-farm income is negative (18.5% lower in the Poisson specification) but statistically insignificant. Fifth, only the promotional session increases students' studying of agriculture as an elective subject, by 11.7 percentage points (23% from 50% in control schools).

While the annual farmer field day, which is organized at the school level, has quantitatively smaller effects on technology adoption, dropouts, and overall school attendance, it is nonetheless an important supplement in making SBAE complete (Panel B of Figure 9). This can be seen in several ways: (i) it speeds up students' adoption of technologies — its effects on the student parcel are positive throughout but statistically significant only for the number of techniques used in the midline (an increase of 0.21, 14% from 1.43 in control schools), and it has no effect on the parent parcel; (ii) it is crucial in ensuring the overturn of SBAE effects on absenteeism and studying hours — reducing endline absenteeism by 42% (a drop of 0.28 from 0.65 out of 5 days) and increasing endline studying hours by 20.8% (1.12 hours from 5.39 hours per week), while together with the promotional session it reverses the SBAE-induced increase in absenteeism (although some marginal effects are not statistically significant, the joint tests reject the null that applying both types of parental engagement has no effects on absenteeism relative to SBAE-only schools, with p = 0.06 in the midline and p = 0.01 in the endline); (iii) it ensures that students' farming hours do not increase; (iv) it is crucial in ensuring positive effects on students' savings, increasing total savings by 23% in the Poisson specification; and (v) only it increases the fraction of students who save for attending university, by 7.5 percentage points (44% from 17% in control schools).

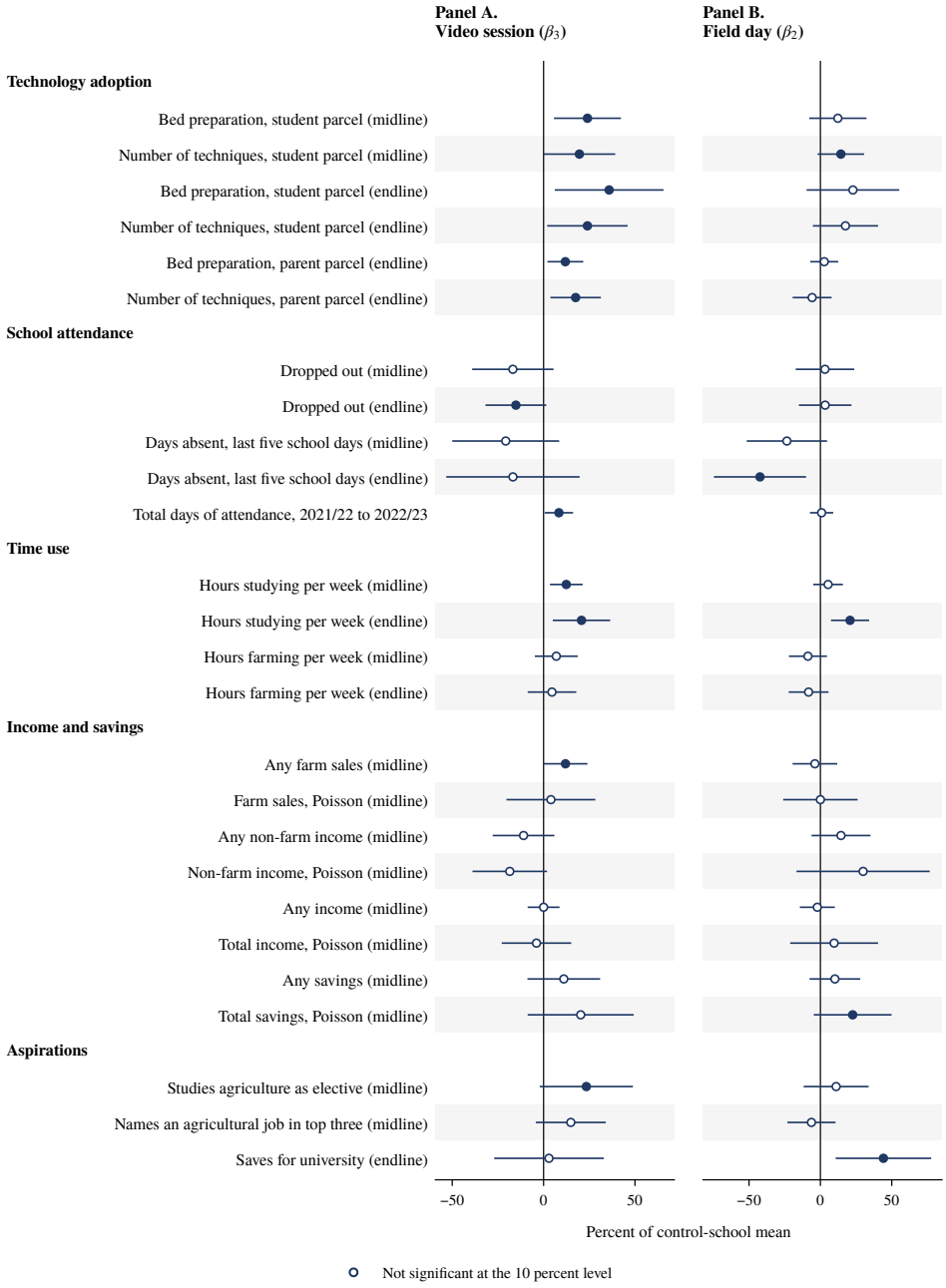


FIGURE 9. MARGINAL EFFECTS OF THE PROMOTIONAL VIDEO SESSION AND THE FARMER FIELD DAY

Notes: This figure plots the marginal effects of the two parental engagement treatments estimated from (1b): β_3 (SBAE × Session; Panel A) and β_2 (SBAE × FFD; Panel B). Because (1b) is the short model, each marginal effect averages over SBAE schools with and without the other treatment. Estimates and 95% confidence intervals are divided by the control-school mean; "dropped out" is one minus the enrollment indicator. Poisson estimates are proportional changes and are shown without rescaling. Standard errors are clustered at the geographical-cluster level. Filled markers denote estimates significant at the 10% level. Lee bounds are reported for the treatment combinations in Figures 4 to 8; full estimates are in Online Appendix Tables OA4 to OA10.

5 Cost-benefit and cost-effectiveness analyses

The previous sections show that SBAE, when augmented with both parental engagement treatments, is highly effective in increasing students' savings, school attendance, and parents' adoption of agricultural technologies. But do these benefits justify the costs and is the augmented SBAE program cost-effective? This section summarizes back-of-the-envelope calculations; the appendix on cost-benefit and cost-effectiveness analyses details all assumptions (Appendix Table B1) and presents the full results (Appendix Figures B1 to B3).

Costs. Over the three-year trial, the augmented SBAE program costs $801,139 (2023 US dollars), or about $40 per student: $621,139 for program implementation and $180,000 for parental engagement. Because setup costs are not repeated and monitoring visits can be curtailed to once a year while teacher training is retained, total costs drop by 63% after the initial three years.

Cost-benefit analysis. I monetize three benefits: the increase in students' savings; students' discounted lifetime earnings from additional days of school attendance ($1.89 per day, assuming an 8% annual return to schooling and a 10% discount rate); and parents' net production value from adopting bed preparation, calculated conservatively. Over the first three years, benefits exceed costs, yielding a benefit-cost ratio of 1.33 (Appendix Figure B1). Even when excluding benefits from parents' technology adoption, the benefit-cost ratio remains at 0.98 — the benefits accruing to students alone nearly justify the program. Over a ten-year horizon, the cumulative benefit-cost ratio is 2.36 if program benefits remain unchanged after the first three years, and 1.53 if they decline to 50% (Appendix Figure B2).

Cost-effectiveness analysis. I compare SBAE's impact per $100 with other African programs evaluated using randomized trials (Appendix Figure B3). For technology adoption, SBAE induces 0.46 to 0.94 households to adopt bed preparation per $100, or 0.55 to 1.02 households to adopt bed preparation *and* a complementary soil management technique, depending on spillovers to non-parent farmers. This far exceeds extension programs using the contact farmer model (0.11 households per $100 in Kondylis et al., 2017) and farmer field days promoting fertilizer (0.15 to 0.46 in Fabregas et al., 2023). It is comparable to video-mediated extension (0.62 in Van Campenhout et al., 2021) and to village-based extension with performance incentives for contact farmers when promoting pit planting (0.60 in BenYishay and Mobarak, 2019), which outperforms SBAE only when promoting composting (1.53). For school attendance, the augmented SBAE program generates 43.46 additional days per $100 — 1.7 to 2.5 times as cost-effective as conditional cash transfers in Malawi (17.06 to 24.98 days per $100 in Baird et al., 2011).

Overall, these back-of-the-envelope calculations provide strong support of the merits of the augmented SBAE program, at least in the context of Liberia. First, SBAE is highly sustainable — it offers a stream of benefits that increasingly outweigh costs after external support for schools is dropped. Second, in terms of cost-effectiveness, SBAE compares favorably with major alternatives in diffusing agricultural technologies and increasing school attendance. The appendix discusses how these results may generalize to countries higher up the ladder of economic development.

6 Mechanisms

Section 4 provides evidence that the SBAE program only succeeds in achieving its goals when parental engagement treatments are included. This section addresses the question of why. I begin by noting that sampled students have baseline farming experiences but may not have participated in the program. So a natural guess would be: parental engagement works by raising students' responses to training in the SBAE program and thus the ITT effects we observe in Section 4.

6.1 SBAE's training effects are not driven by parental engagement

I first document SBAE's training effects on teachers and students, and then show that parental engagement does not drive them.

6.1.1 SBAE impact on teachers

To show that the program is effective in training teachers, we begin by documenting SBAE's impact on teachers, which was measured during the midline and endline surveys. I run the following regression: where Y_{tscp} is the outcome of interest for teacher t, school s, geographical cluster c, and pair-of-clusters p; SBAEp indicates whether pair p is randomly assigned to the SBAE arm; γ_p denotes randomization pair fixed effects; ε_{tscp} is an error term. Standard errors are clustered at the geographical cluster level, which is the highest level at which random assignment took place. β_1 is the coefficient of interest, capturing the ITT effects of SBAE on teachers.

$$Y_{tscp} = \alpha + \beta_1 \text{SBAE}_p + \gamma_p + \epsilon_{tscp} \tag{2}$$

Figure 10 presents the results, with full estimates in Online Appendix Table OA11. The first four estimates indicate consistently positive ITT effects of SBAE on teachers' knowledge about promoted techniques and beliefs about students' attainable yields, in both the midline and endline surveys. Teachers' scores on knowledge tests improve by 12% (0.4σ) and 14% (0.44σ) respectively. Teachers' beliefs about students' attainable cassava tubers in 9-10 months improve by 24% (0.53σ) and 20% (0.42σ) respectively. The large magnitude of these ITT effects validates both (i) the strong receptiveness among teachers and (ii) the novelty of SBAE's agricultural curriculum, considering that 67 out of 97 control schools also have school gardens and most teachers farm outside schools. The last two estimates indicate also positive ITT effects of SBAE on teachers' pedagogy in the endline survey.[18] 12.6 percentage points more students report that their teachers have used agricultural projects or field visits in instruction in the last five days. Students also report a 25% (0.23σ) increase in the number of scientific concepts that teachers have ever demonstrated with farm work. Taken together, Figure 10 thus indicates that the SBAE program has strong effects on teachers as intended.

6.1.2 Early training effects on students

Now we analyze SBAE's impact on students. We first discuss the results of the first follow-up survey, which captured training effects by 5-7 months, and importantly, before parental engagement treatments were implemented. I run the following regression: where Y_{hscp} is the outcome of interest for household h, school s, geographical cluster c, and pair-of-clusters p; SBAEp indicates whether pair p is randomly assigned to the SBAE arm; γ_p denotes randomization pair fixed effects; ε_{hscp} is an error term. Standard errors are clustered at the geographical cluster level, which is the highest level at which random assignment took place. β_1 is the coefficient of interest, capturing the ITT effects of SBAE on students.

$$Y_{hscp} = \alpha + \beta_1 \text{SBAE}_p + \gamma_p + \epsilon_{hscp} \tag{3}$$

Figure 11 presents the results, with full estimates in Online Appendix Table OA12. The first four estimates indicate consistently positive ITT effects of SBAE on students' exposure to promoted farming techniques. The fractions of students who have ever used bed preparation and composting improve by 4 and 6 percentage points respectively. Reflecting SBAE's training on farm management skills, students' knowledge about site

[18] In the Online Appendix, I show similar positive ITT effects on teachers' pedagogy in the midline survey. The results are also robust to using teacher reports.

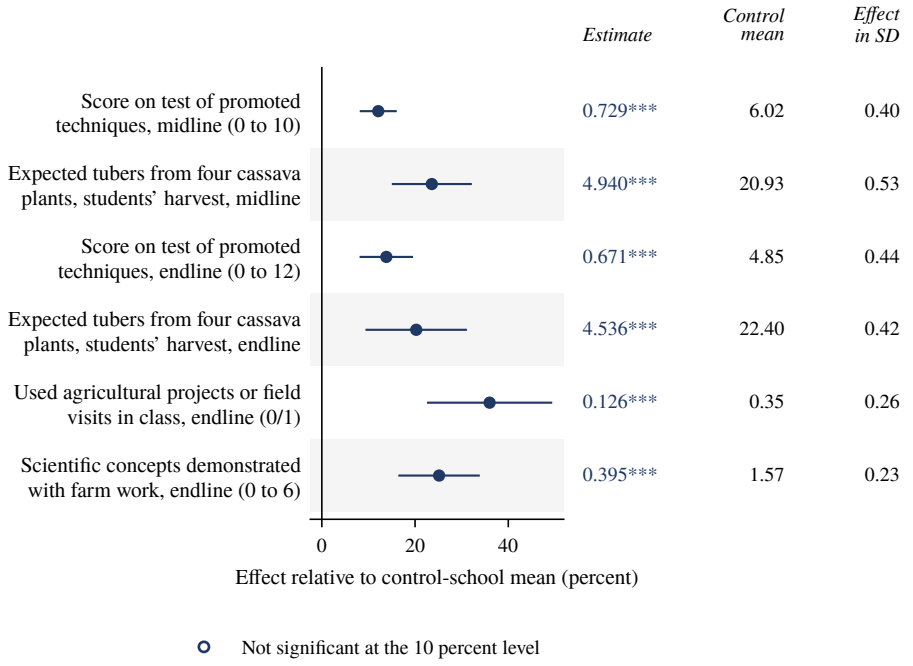


FIGURE 10. MIDLINE AND ENDLINE INTENT-TO-TREAT EFFECTS ON TEACHERS

Notes: This figure plots intent-to-treat effects of SBAE on agriculture and science teachers, estimated from (2). The left panel plots each estimate and its 95% confidence interval divided by the control-school mean; hollow markers denote estimates that are not significant at the 10% level. The right panel reports the estimate in original units, the control-school mean, and the estimate divided by the control-group standard deviation. All regressions include pair-of-clusters fixed effects; standard errors are clustered at the geographical-cluster level. Test scores are on promoted agricultural techniques; expected tubers are teachers' beliefs about the number of tubers students can harvest in 9–10 months from 4 cassava plants. The two pedagogy outcomes are student reports: whether teachers used agricultural projects or field visits in science instruction in the past five days, and the number of scientific concepts teachers have demonstrated using farm work. Full estimates are in Online Appendix Table OA11. Significance: *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

selection improves by 3% (0.08σ). Most importantly, students' beliefs about students' attainable cassava tubers in 9-10 months improve by 4% (0.09σ) — these early improvements in beliefs are consistent with descriptions in agronomic trials that report early, visible differences in the height of stems and the size of leaves when bed preparation is applied. To verify that these changes in beliefs are due to promoted techniques in the SBAE program, the fifth estimate shows that there are no changes in students' beliefs about cassava tubers attained by typical farmers in their communities. The last three estimates indicate ITT effects on students' behavior outside schools. The fraction of students who report intending to look after a farm outside school in the first rainy season improves by 9 percentage points from the control mean of 60%. Meanwhile, there are early signs of pitfalls as discussed in the Introduction — students' weekly studying hours drop by 5% and farming hours increase by 2%, though the latter is not statistically significant at conventional levels. Taken together, Figure 11 thus indicates that before parental engagement, the SBAE program had already increased students' intentions to start home projects and apply promoted techniques outside schools, with some diversion of students' efforts from academic studies.

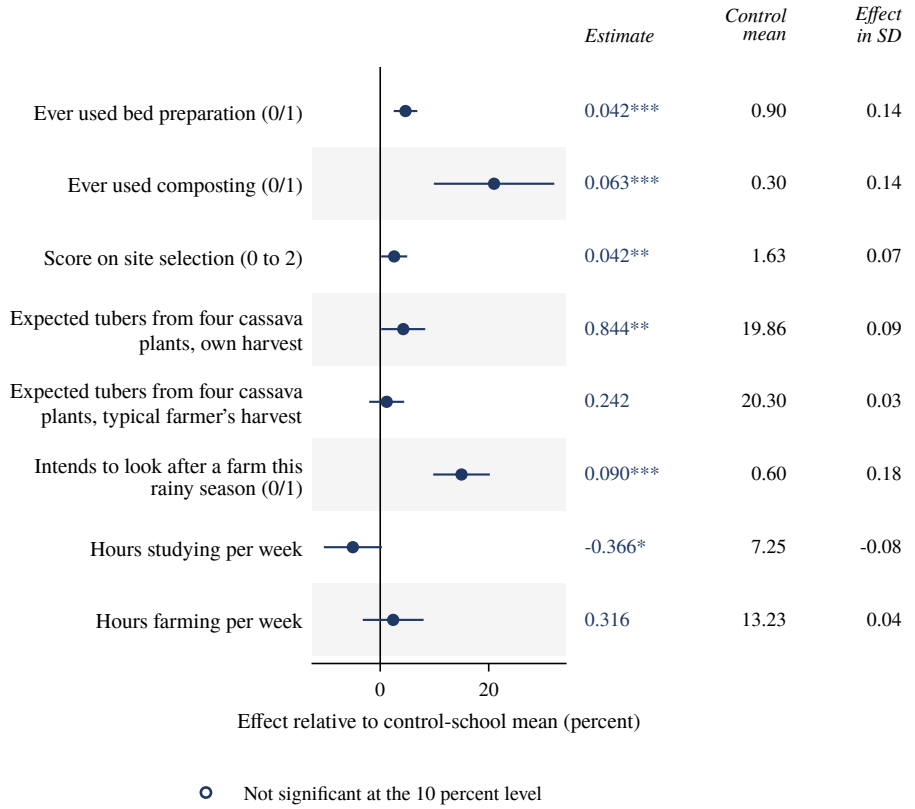


FIGURE 11. INTENT-TO-TREAT EFFECTS ON STUDENTS IN THE FIRST FOLLOW-UP SURVEY

Notes: This figure plots intent-to-treat effects of SBAE on students in the first follow-up survey, before the parental engagement treatments, estimated from (3). The left panel plots each estimate and its 95% confidence interval divided by the control-school mean; hollow markers denote estimates that are not significant at the 10% level. The right panel reports the estimate in original units, the control-school mean, and the estimate divided by the control-group standard deviation. All regressions include pair-of-clusters fixed effects; standard errors are clustered at the geographical-cluster level. Expected tubers are students' beliefs about the number of tubers that they (own harvest) or a typical farmer in their community can harvest in 9–10 months from 4 cassava plants. Hours are for the week preceding the survey. Full estimates are in Online Appendix Table OA12. Significance: *** p<0.01, ** p<0.05, * p<0.10.

6.1.3 Could increased responses to SBAE training explain?

Figure 12 examines this possibility by estimating midline and endline treatment effects on key intermediate outcomes associated with students' training in the SBAE program; full estimates are in Online Appendix Table OA13. Mirroring results in the first follow-up survey (Section 6.1.2), the odd-numbered columns of Online Appendix Table OA13 indicate that the overall SBAE effect is positive as expected — 73.7% and 84.3% of sampled students in SBAE schools had participated in the program by the midline and endline surveys; SBAE school students scored 6.8% and 5.7% higher than control school students on tests of promoted techniques in the midline and endline; 5.8 and 7.9 percentage points more students in SBAE schools believed that they could harvest more cassava tubers per plant than a typical adult farmer in the community.

A closer inspection of Figure 12 suggests that the majority of student participation in the program is driven by SBAE alone. Specifically, 67.7% of sampled students in SBAE-only schools had participated by the midline, and the program promotional session added only 13.2% of students to the program. By the endline, 80.9% of sampled students in SBAE-only schools had participated, whereas the program promotional session had no statistically significant effects on program participation. The evidence thus suggests that

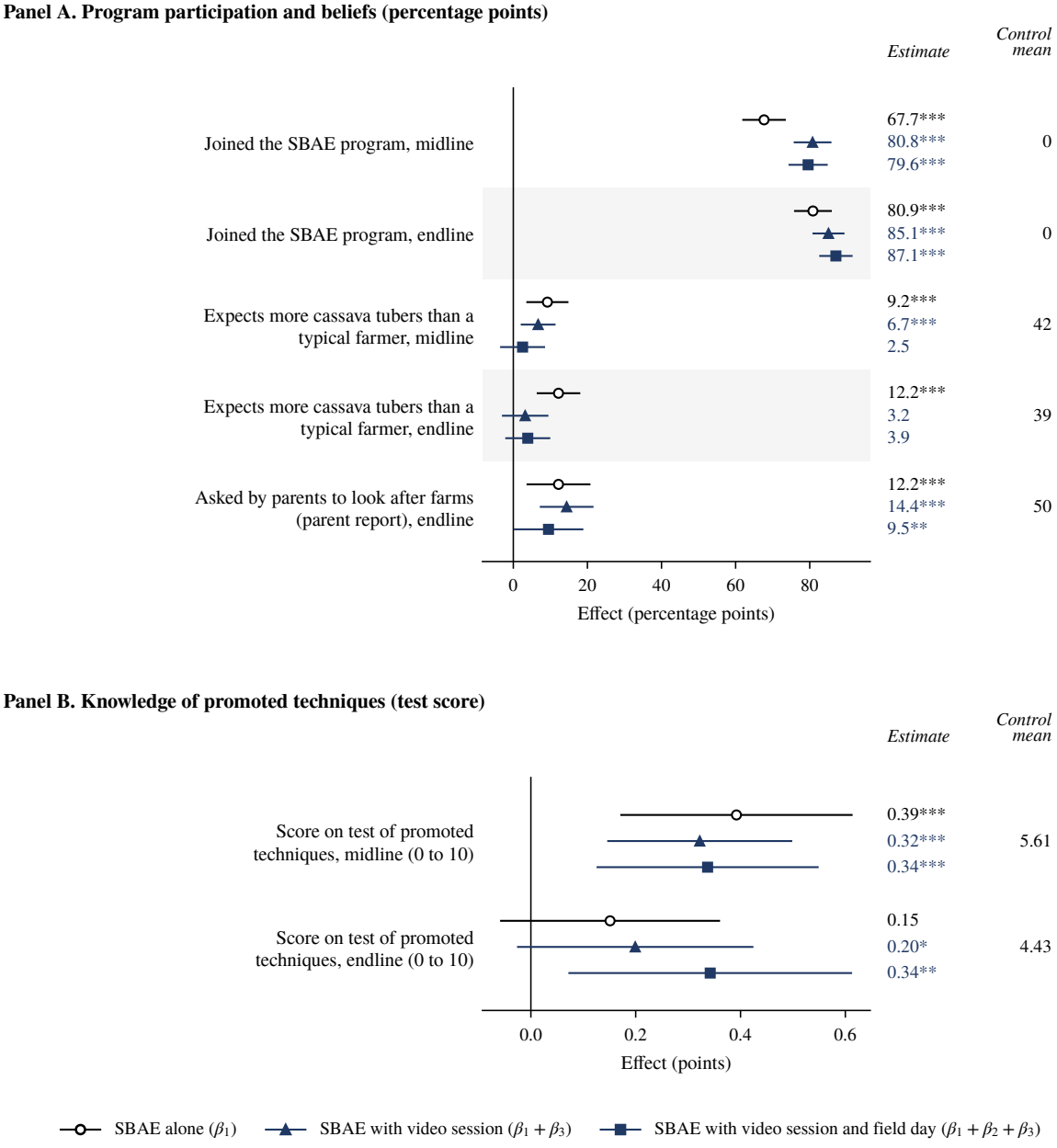


FIGURE 12. MIDLINE AND ENDLINE INTENT-TO-TREAT EFFECTS ON STUDENTS' TRAINING OUTCOMES

Notes: This figure plots intent-to-treat effects estimated from (1b) for SBAE alone (β_1; hollow circles), SBAE with the program promotional session ($\beta_1+\beta_3$; triangles), and SBAE with both the promotional session and the annual farmer field day ($\beta_1+\beta_2+\beta_3$; squares). Panel A reports indicator outcomes in percentage points; Panel B reports scores on the test of promoted techniques (0–10). Horizontal bars are 95% confidence intervals based on standard errors clustered at the geographical-cluster level; hollow markers denote estimates that are not significant at the 10% level. "Joined the SBAE program" is zero in control schools by construction. "Student asked to look after farms" is a parent report from the endline survey; all other outcomes are student reports. Full estimates, including the overall SBAE effect from (1a), the field day combination ($\beta_1 + \beta_2$) and Lee bounds, are in Online Appendix Table OA13. Significance: *** p<0.01, ** p<0.05, * p<0.10.

the program promotional session merely speeds up SBAE participation, and only by a small fraction. For increased SBAE participation to explain the dramatic effects of parental engagement — large differentials on technology adoption and reversed effects on education — we would expect marginal effects of parental engagement on SBAE participation that are large relative to pure SBAE effects.

Of course, parental engagement could have intensified students' absorption of SBAE training, even if participation remains unchanged. However, another closer inspection — of the test scores and beliefs in Figure 12 — suggests that SBAE alone drives the improvements in students' agricultural knowledge (at midline) and beliefs, whereas the parental engagement treatments have no statistically significant positive effects on them. For increased students' absorption of SBAE training to explain the dramatic effects of parental engagement, we would expect that parental engagement treatments at least exert some effects on students' agricultural knowledge and beliefs.

So far we have examined measures that capture SBAE's impact within schools. But an important part of SBAE is for students to become agents of diffusion. Could it be that students only increased their intentions to look after household farms upon parental engagement? However, results in the first follow-up survey (Section 6.1.2) suggest that SBAE increased students' intentions to look after household farms before parental engagement. The last indicator in Panel A of Figure 12 further examines this from endline parent reports — whether students have asked to look after farms in the last two years. SBAE alone increases this measure by 12.2 percentage points (from 50% in control schools), whereas the program promotional session and the annual farmer field day have no statistically significant effects.

To sum up, the evidence suggests that the pivotal effects of parental engagement on primary outcomes cannot be driven by students' program participation, absorption of SBAE training, or intentions to look after household farms — SBAE alone accounts for these changes.

6.2 Parental engagement enables intra-household diffusion mechanisms otherwise muted

Having shown that SBAE alone is sufficient to equip students as agents of diffusion, we now turn to another possibility: parental engagement enables intra-household diffusion mechanisms otherwise muted, despite students' willingness to apply promoted techniques. I focus specifically on the dual diffusion mechanisms mentioned in the Introduction: (i) parents' visits to schools; (ii) students' home projects, intra-household communication, and parents' observations.

Parents' visits to schools. Figure 13 shows results on parents' visits to schools and related outcomes, with full estimates in Online Appendix Table OA14. Columns (1) and (3) of that table indicate that, in line with a larger fraction of SBAE schools (100%) having school farms than control schools (69%), SBAE overall has increased the fraction of parents who have visited school farms and the number of visits parents make in the second rainy season.

A closer inspection, however, suggests that parents' visits to schools farms are far from mechanical. First, only a small fraction of SBAE-only school parents visit school farms — 43.9%, as opposed to 33% in control schools — whereas including both types of parental engagement into the SBAE program increases this fraction to 65.3%. Second, and most surprisingly, SBAE alone does not increase the number of visits that parents make in the second rainy season. Including both types of parental engagement increases this measure by 0.94, which means doubling visits from control schools (0.89 per parent). Third, and most importantly, including both types of parental engagement is crucial in enabling more parents to observe harvests on the school farm, from 37.9% in SBAE-only schools to 61.1%.

Besides visits to school farms, Figure 13 also examines parents' attendance of parent-teacher meetings, which could be crucial in encouraging parents' monitoring of students' academic studies and school attendance. In line with treatment effects on students' absenteeism and studying hours (Section 4.2.2), Figure 13 indicates a reversal of effects on parents' attendance of parent-teacher meetings. With SBAE only, parents' interest in attendance surprisingly drops — 10.2 percentage points fewer parents attend meetings (from 36%

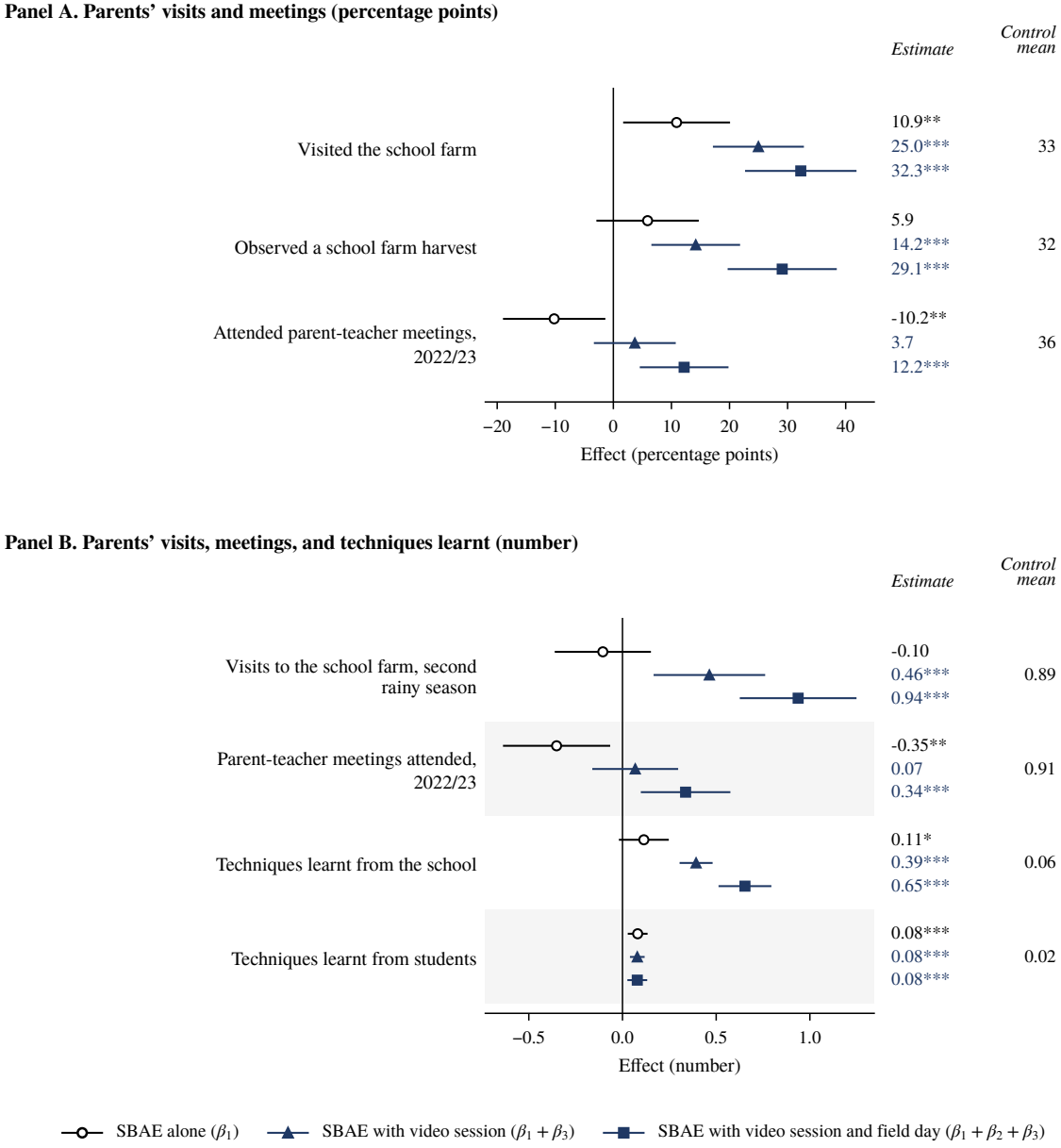


FIGURE 13. ENDLINE INTENT-TO-TREAT EFFECTS ON PARENTS' VISITS TO SCHOOLS AND RELATED OUTCOMES

Notes: This figure plots intent-to-treat effects estimated from (1b) for SBAE alone (β_1; hollow circles), SBAE with the program promotional session ($\beta_1+\beta_3$; triangles), and SBAE with both the promotional session and the annual farmer field day ($\beta_1+\beta_2+\beta_3$; squares). Panel A reports indicator outcomes in percentage points; Panel B reports counts. Horizontal bars are 95% confidence intervals based on standard errors clustered at the geographical-cluster level; hollow markers denote estimates that are not significant at the 10% level. All outcomes are endline parent reports. Visits are counted for the second rainy season (2022) and parent-teacher meetings for the 2022/23 school year; techniques learnt are promoted techniques that parents report having first learnt from the school or from students in the past two years. Full estimates, including the overall SBAE effect from (1a), the field day combination (β_1 + β_2) and Lee bounds, are in Online Appendix Table OA14. Significance: *** p<0.01, ** p<0.05, * p<0.10.

in control schools) and the number of meetings attended in 2022/23 drop by 38.7% (a drop of 0.35 from 0.91 in control schools). Both the program promotional session and the annual farmer field day contribute to a reversal in attendance — when included together in the SBAE program, 12.2 percentage points more parents attend (relative to 36% in control schools) and the number of meetings attended increase by 37%. These results indicate that, in line with potential concerns about SBAE, the program alone could distract parents' attention from students' academics — whereas the inclusion of parental engagement helps to reverse this distraction.

Is there further evidence that substantiates the impact of parents' visits to schools? The last two rows of Panel B of Figure 13 examine if these visits increase parents' learning of promoted techniques. Specifically, I assess channels through which parents learn promoted techniques. In the endline survey, after probing parents' awareness of promoted techniques in open-ended questions about planting methods, enumerators follow up by asking when and how they first learnt the promoted techniques — in particular, whether they learnt it from the SBAE schools, from students specifically, or from other members of the community. These rows examine treatment effects on the number of techniques parents have learnt, focusing on reports of learning within the past two years. Two interesting patterns emerge. First, SBAE alone increases parents' learning from both schools (0.114 techniques; 190% rise from control schools) and students (0.081 techniques; 405% from control schools). Second, both the program promotional session and the annual farmer field day substantially increase parents' learning from schools, but not from students. Specifically, including both types of parental engagement increases the number of techniques parents learn from schools from 0.174 to 0.714 — an increase by around fourfold. This pattern suggests that parental engagement is particularly important in enabling parents to learn the technical aspect of promoted techniques via visits to schools — a function that is less fulfilled by intra-household interactions.

Students' home projects, communication with parents, and parents' observations. Figure 14 turns to the second diffusion mechanism of SBAE — intra-household interactions between students and parents, with full estimates in Online Appendix Table OA15. Panel A examines whether SBAE leads to increased student home projects, as manifested in various proxies about whether students manage household farms. Its first four rows focus on two measures that are reported by students in the midline and endline surveys: (i) whether students look after farms; (ii) whether students choose planting methods on the student parcel. Two patterns emerge. First, while SBAE has some positive but imprecise effects on these proxies of student-led farms, the program promotional session intensifies the process. Specifically, SBAE alone raises the fraction of students who look after farms in the midline and endline — only slightly — to 56.3% (from 53%) and 54.5% (from 48%); with the program promotional session, these measures increase to 63.6% and 58.8% respectively. The contrast is even sharper if we focus on the more stringent proxy about student-led farms — whether they choose planting methods on the student parcel. SBAE alone raises this fraction in the midline and endline to 23.9% (from 18%) and 27% (from 23%); with the program promotional session, these measures increase to 30.8% and 34.9%.

Besides student-led home projects, students could also influence household technology adoption and educational decisions through their communication with parents and parents' observations. The remaining rows turn to these other mechanisms of intra-household interactions, examining whether parental engagement has encouraged more active communication between students and parents. The fifth row of Panel A examines the fraction of students who have asked parents for permission to change farming practices on household farms, as reported by parents. Both the program promotional session and the annual farmer field day significantly increase this fraction. Their collective impact is substantial, increasing this fraction from 41.3% in SBAE-only schools (37% in control schools) to 56.3%.

Panel B examines three measures of communication that are reported in the midline student survey. These questions were asked as follow-ups when students have adopted promoted techniques on the student parcel. The three measures are: the number of techniques (i) that students have shown parents how to make; (ii) that students have shown parents the associated harvests; (iii) that students have advised parents to use on

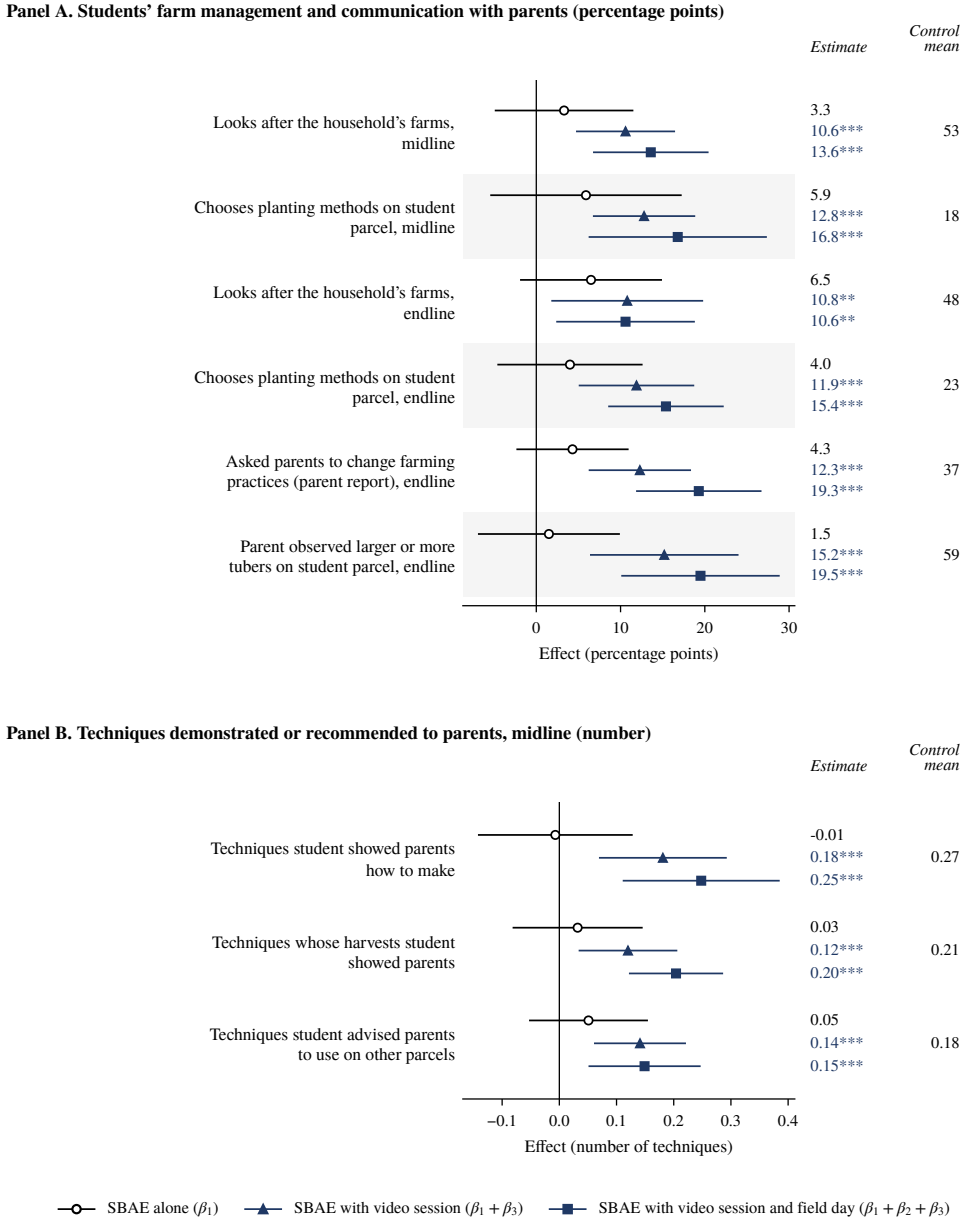


FIGURE 14. MIDLINE AND ENDLINE INTENT-TO-TREAT EFFECTS ON STUDENTS' MANAGEMENT OF HOUSEHOLD FARMS AND RELATED OUTCOMES

Notes: This figure plots intent-to-treat effects estimated from (1b) for SBAE alone (β_1; hollow circles), SBAE with the program promotional session ($\beta_1+\beta_3$; triangles), and SBAE with both the promotional session and the annual farmer field day ($\beta_1+\beta_2+\beta_3$; squares). Panel A reports indicator outcomes in percentage points; Panel B reports numbers of promoted techniques. Horizontal bars are 95% confidence intervals based on standard errors clustered at the geographical-cluster level; hollow markers denote estimates that are not significant at the 10% level. Looking after farms, choosing planting methods and the three technique counts are student reports; asking parents to change farming practices and observing larger or more tubers on the student's parcel are endline parent reports. "Other parcels" are parcels other than the student's own. Full estimates, including the overall SBAE effect from (1a), the field day combination ($\beta_1 + \beta_2$) and Lee bounds, are in Online Appendix Table OA15. Significance: *** p<0.01, ** p<0.05, * p<0.10.

other parcels. Parental engagement substantially raises these measures (i) from 0.26 to 0.52 techniques (0.27 in control schools); (ii) from 0.24 to 0.41 techniques (0.21 in control schools); and (iii) from 0.23 to 0.33 techniques (0.18 in control schools).

Students' home projects, requests to change practices on household farms, and communication with parents can all influence parents' technology adoption via their impact on parents' beliefs about the returns from adoption. The last row of Panel A captures this mechanism by asking endline parents whether they have observed improved root crop harvests on students' parcels in the last two years, where improved root crop harvests is defined as either an increase in the size or number of tubers per plant. Both the program promotional session and the annual farmer field day raise this fraction, though only the former's effect is statistically significant and it is around three times larger. Their collective impact is substantial — raising the fraction from 60.5% (59% in control schools) to 78.5%.

6.3 Standalone effects of SBAE on primary outcomes

Figure 15 plots the standalone effect of SBAE on every primary outcome.

Despite its positive training effects on teachers and students (Section 6.1), SBAE on its own does not improve most of the outcomes it targets. Taking the outcomes in the order of Figure 15: (i) Technology adoption. On the student parcel, SBAE alone has small and statistically insignificant effects on both the use of bed preparation and the number of promoted techniques used, at both midline and endline. On the parent parcel, it has essentially zero effect on the use of bed preparation, although it increases the number of techniques used by 0.2 (14% from 1.38 in control schools). (ii) School attendance. SBAE alone has no effect on dropouts but increases absenteeism by 26.2% in the midline (an increase of 0.18 from 0.69 out of 5 days) and 31.2% in the endline (an increase of 0.2 from 0.65 out of 5 days); both increases survive the Lee bounds. Its effect on total days of attendance is small and statistically insignificant. (iii) Time use. SBAE alone reduces students' weekly studying hours by 10.6% in the midline (a drop of 0.6 hours from 5.69 hours per week) and by 19.9% in the endline (a drop of 1.07 hours from 5.39 hours per week), consistent with the drop in students' studying hours in the first follow-up survey, while leaving farming hours unchanged. (iv) Income and savings. SBAE alone has no significant effect on farm sales, reduces the fraction of students with non-farm income by 6.1 percentage points (13% from 47% in control schools), and has no significant effect on total income or savings. (v) Aspirations. SBAE alone has no significant effect on whether students study agriculture as an elective subject, increases students' naming of agricultural careers by 7.8 percentage points (18% from 43% in control schools), and has no significant effect on whether students save for attending university.

Pooling all SBAE schools, as in (1a), masks these standalone effects. The overall SBAE effects, reported in the odd-numbered columns of Online Appendix Tables OA4 to OA10, are positive for technology adoption on both parcels, enrollment, farm sales, the fraction of students with savings, and all three measures of aspirations, and close to zero for absenteeism and studying hours, because they average over SBAE schools with and without parental engagement.

6.4 Summary

I have presented evidence that parental engagement effects are unlikely due to increased student participation or responses to SBAE training in Section 6.1.3. This implies that the pivotal effects of parental engagement on the primary outcomes must be due to increased impact outside schools. Section 6.2 presents solid evidence that the dual intra-household mechanisms of SBAE — parents' visits to schools and intra-household interactions between students and parents — are enabled by parental engagement. Remarkably, the SBAE program on its own has only small, and in several cases negative, effects on these mechanisms. Further, each mechanism has distinct effects on parents' exposure to promoted techniques — (i) parents' visits to schools lead to a fourfold increase in parents' learning of promoted techniques from schools; (ii) students' home projects and

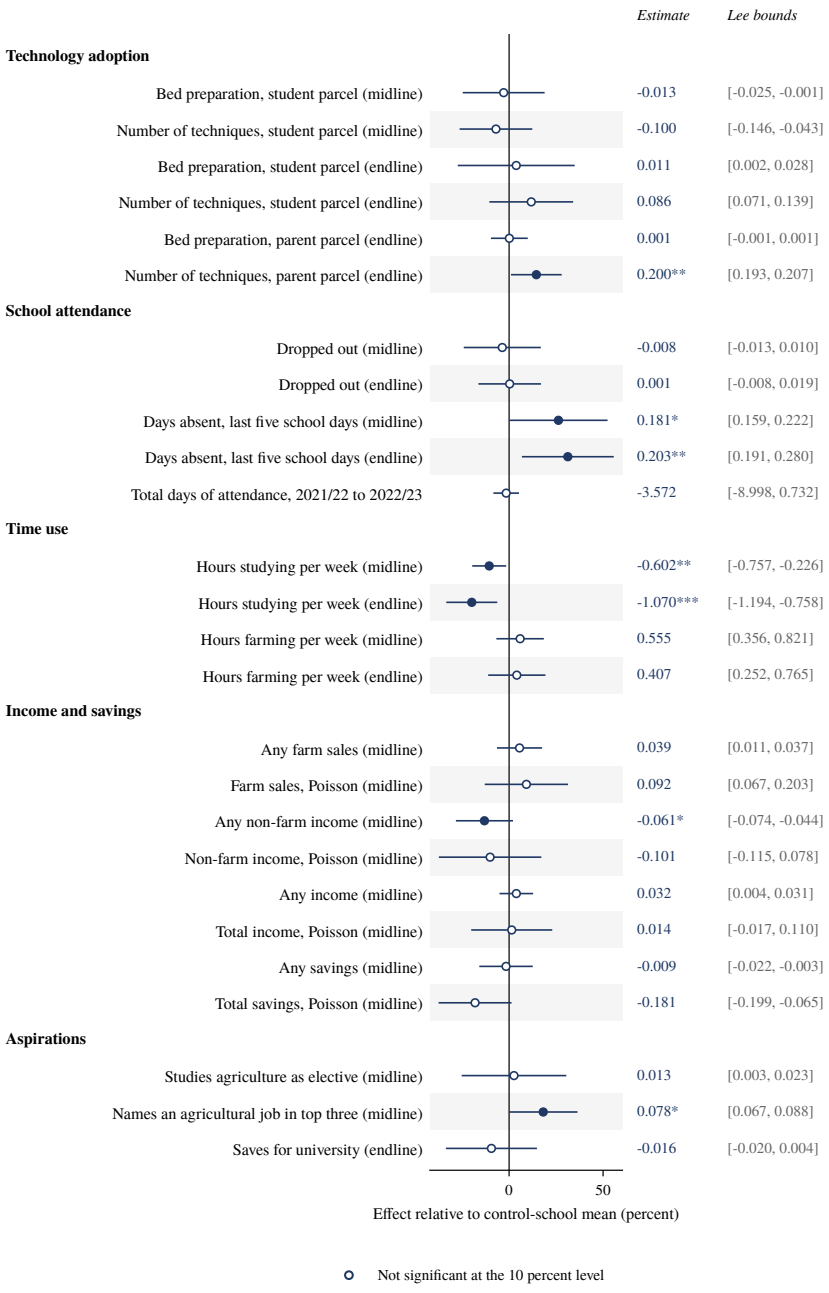


FIGURE 15. STANDALONE EFFECTS OF SBAE WITHOUT PARENTAL ENGAGEMENT

Notes: This figure plots the intent-to-treat effect of SBAE without parental engagement (β_1 in (1b)) on each primary outcome. In the left panel, estimates, 95% confidence intervals and Lee bounds (brackets) are divided by the control-school mean; Poisson estimates are proportional changes and are shown without rescaling. Filled markers denote estimates significant at the 10% level. The right panel reports the estimate in original units with its Lee bounds (stars on each bound are based on its own standard error) and the estimate relative to the control-school mean. "Dropped out" is one minus the enrollment indicator, so its estimates and bounds have the opposite sign to those in Online Appendix Table OA6. Standard errors are clustered at the geographical-cluster level. Significance: *** p<0.01, ** p<0.05, * p<0.10. Full estimates are in Online Appendix Tables OA4 to OA10.

increased interactions with parents leads to a 13-fold increase in the treatment effect on parents' observations of improved yields on students' parcels. Both channels could contribute to improvements in technology adoption and school attendance via increasing parents' increased acceptance of promoted agricultural techniques.

Online Appendix Table OA16 presents evidence that parental engagement indeed produces downstream effects on parents' agricultural knowledge and beliefs, as well as changes in attitudes towards education. Column (2) indicates that SBAE, on its own, increases parents' score on the test of promoted techniques by only 0.12 (from 5.23 out of 12 in control schools), which is statistically insignificant; whereas the treatment effect rises to 0.352 (6.7% increase from 5.23 out of 12 in control schools) when both types of parental engagement are included. Column (4) indicates that SBAE alone increases parents who believe that students would harvest more cassava tubers per plant than a typical adult farmer in the community by 4.4 percentage points (from 35% in control schools), which is statistically insignificant; this treatment effect increases to 10.5 percentage points with parental engagement. Lastly, mirroring the annual farmer field day's effects on absenteeism as well as studying and farming hours (Online Appendix Tables OA6 and OA7), column (6) shows that it reduces the fraction of students who report having missed school in 2022/23 due to requests from household members to work by 7.5 percentage points (from 24% in control schools).

The evidence on mechanisms also contributes to our discussion of the relative importance of the program promotional session and the annual farmer field day, as in Section 4.3. Both types of parental engagement are quantitatively important in enabling the first diffusion mechanism — parents visiting schools and attending parent-teacher meetings, learning promoted techniques, and observing harvests on school farms. But only the program promotional session enables the second diffusion mechanism — students starting home projects and triggering parents' observation of improved yields on household farms. These differences likely explain why increased parents' technology adoption, increased students' farm sales, and reduced dropouts are only observed when the program promotional session is included. Meanwhile, the annual farmer field day supplements by reducing household requests of students to work, which is crucial in reducing absenteeism and overall child labor.

6.5 Contrasting the marginal effects of the video and the farmer field day

Contrasting the marginal effects of the farmer field day and the promotional video sheds light on what drives households' technology adoption and the reversal of SBAE effects on students' school attendance. Both interventions provide information to parents, but of different types. The farmer field day offers hands-on technical training on promoted soil management techniques, which parents can employ while retaining students as assistants on family farms. The promotional video, by contrast, highlights how SBAE creates complementarities between students' agricultural work and education—emphasizing that students can manage farms independently and pursue further education to strengthen their skills as farmers. If the key constraint to technology adoption is parents' technical knowledge or program awareness, the farmer field day should have qualitatively similar effects as the video. If instead the constraint is parents' beliefs about the value of delegating farms to students and, more broadly, the returns to supporting students' education, the video should prove more effective—particularly for activating intrahousehold technology transfer—the second diffusion mechanism of SBAE.

Figure 16 presents the marginal effects on SBAE's first diffusion mechanism: parents' visits to school farms. Both the farmer field day and the promotional video play important roles in activating this channel. The farmer field day increases the fraction of parents who visit school farms by 7.3 percentage points and the number of visits in the second season by 0.47 (59% from the SBAE mean of 0.8); the video produces stronger, but qualitatively similar effects of 14.1 percentage points and 0.57, respectively. Similarly, both treatments increase the fraction of parents who observe school farm harvests (by 15 and 8.3 percentage points, respectively), the number of techniques parents learn from school (by 0.26 and 0.28, 150% and 160% from the SBAE mean of 0.17), and attendance at parent-teacher meetings (by 0.27 and 0.42, 48% and 75% from the

SBAE mean of 0.56) in the third school year. Overall, therefore, the evidence suggests that both treatments are effective in encouraging parents to visit schools, learn promoted techniques, and monitor students' academic progress.

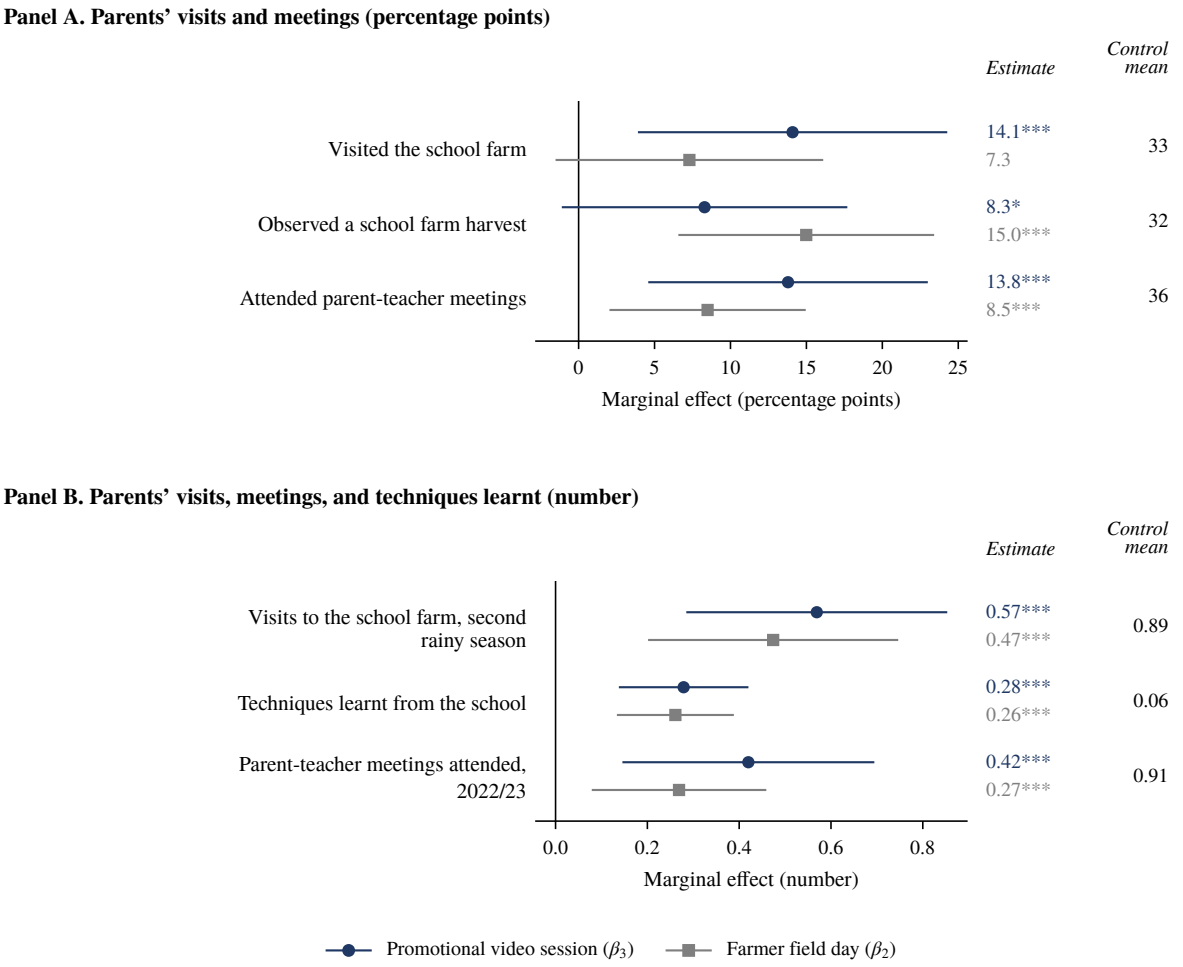


FIGURE 16. CONTRASTING MARGINAL TREATMENT EFFECTS ON PARENTS' VISITS TO SCHOOLS

Notes: This figure plots the marginal effects of the program promotional session (β_3 in (1b); dark blue circles) and the annual farmer field day (β_2; gray squares). Panel A reports indicator outcomes in percentage points; Panel B reports counts. Horizontal bars are 95% confidence intervals based on standard errors clustered at the geographical-cluster level; hollow markers denote estimates that are not significant at the 10% level. All outcomes are from the endline parent survey. Visits are counted for the second rainy season (2022) and parent-teacher meetings for the 2022/23 school year. Full estimates, including β_1 and $\beta_1 + \beta_2 + \beta_3$, are in the even-numbered columns of Online Appendix Table OA14. Significance: *** p<0.01, ** p<0.05, * p<0.10.

Figure 17 presents evidence on the second diffusion mechanism—intra-household technology diffusion from students to parents. Here, the video's dominance is stark. Only the video, not the farmer field day, increases the fraction of students who look after household farms. The video increases the fraction of students who look after farms by 7.3 percentage points at midline (significant at the 10% level) and 4.3 percentage points at endline (statistically insignificant), while the corresponding farmer field day effects are negligible (3 percentage points and -0.2 percentage points, respectively). More strikingly, the video's effect on the number of promoted techniques students demonstrate to their parents is approximately three times that of the farmer field day, and its effect on the number of techniques students advise parents to adopt on their own parcels is

over ten times as large. In turn, the increase in the fraction of parents who report observing improved cassava yields on students' parcels is substantially higher with the video (13.7 percentage points) than with the farmer field day (4.3 percentage points)—a ratio of more than three to one. Overall, therefore, the evidence suggests that the video, not the farmer field day, plays a dominant role in activating the second diffusion mechanism of SBAE.

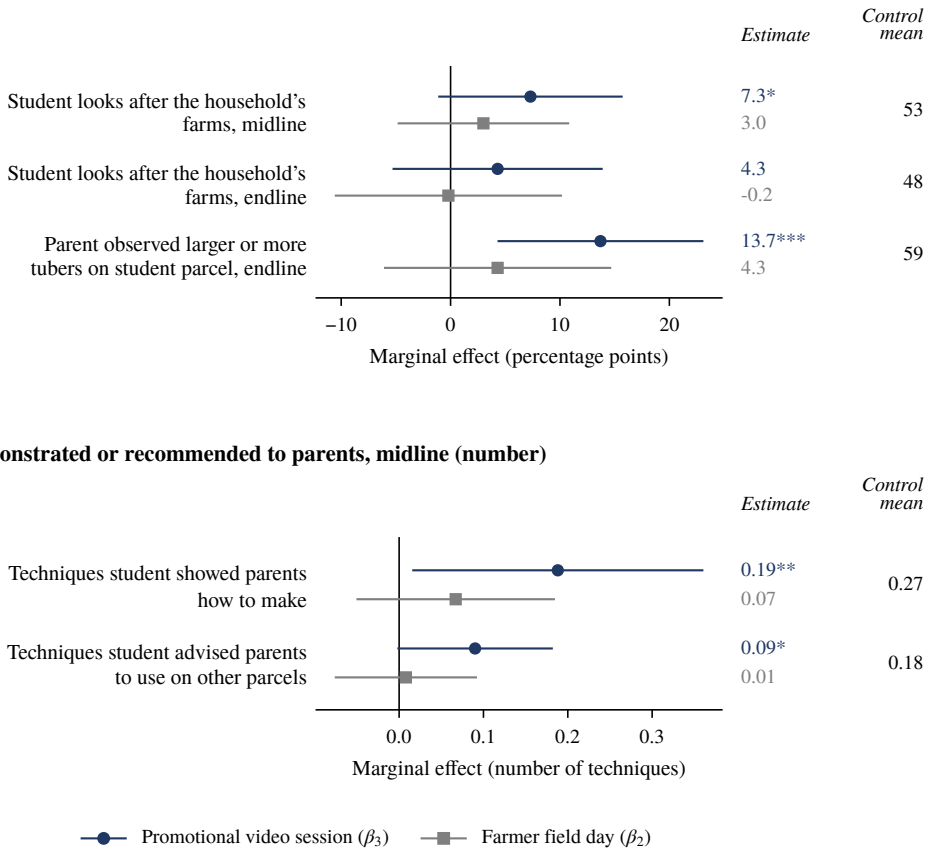


FIGURE 17. CONTRASTING MARGINAL TREATMENT EFFECTS ON INTRAHOUSEHOLD TECHNOLOGY DIFFUSION

Notes: This figure plots the marginal effects of the program promotional session (β_3 in (1b); dark blue circles) and the annual farmer field day (β_2; gray squares). Panel A reports indicator outcomes in percentage points; Panel B reports counts. Horizontal bars are 95% confidence intervals based on standard errors clustered at the geographical-cluster level; hollow markers denote estimates that are not significant at the 10% level. Looking after the household's farms and the numbers of techniques shown and advised are student reports; the observation of larger or more tubers on the student's parcel is a parent report from the endline. "Other parcels" are parcels other than the student's own. Full estimates, including β_1 and $\beta_1 + \beta_2 + \beta_3$, are in the even-numbered columns of Online Appendix Table OA15. Significance: *** p<0.01, ** p<0.05, * p<0.10.

As highlighted in the introduction, since farmers prefer learning from socioeconomic peers (BenYishay and Mobarak, 2019), observing technologies under familiar conditions (Munshi, 2004), and gathering multiple observations before adopting (Beaman et al., 2021), students' home projects—SBAE's second diffusion mechanism—could prove pivotal in overcoming barriers for technology adoption. Consistent with this, Figure 9 shows that while both treatments contribute to technology adoption on students' parcels, only the video produces effects strong enough to increase adoption on parents' parcels. This has significant implications for households' production, as parents' production tends to be bigger in scale and generate far greater farm sales than students' parcels. Similarly, consistent with the video reshaping parents' aspirations for their children's

education, the video accounts for the entire reduction in student dropouts, and, correspondingly, the entire increase in the total number of days students attend school.

7 A puzzle and its interpretation

7.1 The puzzle

SBAE aims to break the cycle in which low agricultural productivity and household labor needs depress school attendance. It works through two channels: school farms that demonstrate improved techniques to parents, and student home projects that test those techniques on the household's own soils. The parental engagement treatments center on a promotional video that explains how SBAE differs from typical school gardens, which merely employ students as farm assistants. Taken together, the results in Sections 4 to 6 have five features that any explanation of parents' responses needs to account for at the same time.

F1. A reversal in sign. Without parental engagement, SBAE fails to raise adoption of the main yield-improving technique on any household parcel, increases absenteeism, and reduces students' studying hours (Section 6.3). With engagement, SBAE is among the most cost-effective agricultural extension programs evaluated in Africa in diffusing technology, and it is 1.7 to 2.5 times as cost-effective as conditional cash transfers in raising school attendance (Section 4.2 and Section 5).

F2. Avoidance of cheap, available information. Over the three years of the trial, 81% of students in schools without parental engagement joined the program, yet only 44% of their parents visited the school farms. Without the informational treatment, SBAE does not increase the number of parents' visits to school farms in the second rainy season, and it reduces the number of parent-teacher meetings that parents attend by 39% relative to control schools (Section 6.2).

F3. A low-content trigger with a lasting response. A one-time promotional session centered on a video was enough to change the sign of the program's effects, and the reversal persisted until the endline survey.

F4. Breadth. Parents' responses are not limited to their engagement with the program. They also extend to their children's studying hours and school attendance.

F5. Same evidence, opposite responses. Parents in SBAE schools with and without parental engagement face the same program and the same evidence about it, while parents in control schools received no program at all.

7.2 Why the standard explanations fall short

Bayesian updating under a single prior. Visiting the school farm takes some time and effort, but the cost is modest. If parents were uncertain about the value of the program, the information gained from a visit should be worth this cost, and parents should visit more often as the program unfolds rather than less often (F2). For visits to fall, parents' prior beliefs must be concentrated enough that what they observe of SBAE is taken as confirming their existing view. Moreover, the content of the news does not by itself determine whether it is good or bad. The main signal that parents in SBAE-only schools receive is their children's report that part of the school day is spent on a farm. This report is bad news if the program is seen as a typical school garden, but good news if it is seen as training in farm management (F1, F5). Whether parents treat the report as good or bad news therefore depends on how they interpret the program, rather than on the report itself.

Bayesian updating after a negative signal. A stronger version of the updating account adds a fixed cost of visiting. With such a cost, there is a range of beliefs in which acquiring further information is not worthwhile. A sufficiently precise negative signal would move a parent into this range, after which she would stop checking on the program. The informational treatment would move her out of this range, and her own subsequent learning would sustain the change. Under this account, the video does not need to do all of the persuading by itself, so neither the size of the reversal nor its persistence is, on its own, evidence against

Bayesian updating (F2, F3). This is the strongest alternative to the interpretation I propose below. However, three features of the setting are difficult to reconcile with it.

First, parents in control schools received no program. The account therefore requires parents in SBAE schools to conclude that a school that received a farm, teacher training, and materials is less worth engaging with than a school that received nothing (F5). Second, the negative signal would need to be misleading, and misleading in the same direction for many parents. Parents in both arms face the same program and the same evidence, and in schools with parental engagement this evidence is favorable enough to sustain the reversal. For the signal to have driven parents in schools without engagement away from the program, it must have been not only unfavorable but also wrong, and wrong in the same direction for enough parents to shift the average. A shared error of this kind is more naturally explained by how parents interpret the evidence than by the particular signals that they happened to receive (F5). Third, the response is broader than this account predicts. A signal about the quality of one program should change parents' engagement with that program, but it does not explain why children's studying hours and school attendance fall (F4).

An imprecise negative signal does not rescue the account either. With a single prior, the value of a visit depends on how uncertain the parent remains, not on how negative the news was. An imprecise signal leaves the parent more uncertain, and therefore raises the value of visiting (F2). Two qualifications apply. First, if the returns to SBAE depend on how many parents engage with the program, a parent's unfavorable reading may be correct rather than mistaken, and the treatment may coordinate parents on a better outcome. This hypothesis implies that the effects on a parent should depend on how many other parents at the same school were treated. Second, no reduced-form comparison can rule out every explanation that works through the level of beliefs, because any response can be rationalized by some unobserved belief content. Distinguishing these accounts requires variation in how the program is presented while holding the information provided fixed, which I plan to pursue in future work.

Selective attention. In models of selective attention, people fail to learn because they do not notice dimensions that they believe to be irrelevant, and pointing out those dimensions restores learning (Schwartzstein, 2014; Hanna et al., 2014). Applied to this setting, the account implies that parents are unaware of the program's science instruction, the new techniques, or students' home projects. Fieldwork suggests otherwise. Parents are aware of these features, but they lack a way of interpreting them under which agriculture and education reinforce each other. The account also does not explain why the same facts are read as bad news in one arm and as good news in the other (F5), or why parents' responses extend to schooling (F4).

Associative memory. Models of associative memory explain why a cue retrieves past experiences that resemble it (Bordalo et al., 2020, 2022, 2026). These models account well for parents' initial reading of the program. A farm at a school evokes familiar school gardens, which are widely associated across Africa with drudgery that trains students to be obedient farm assistants (FAO, 2009). The models also allow a different cue to retrieve a different category, which fits the immediate effect of the video (F1, F5). However, they do not describe how people search for information. Because they select one representation at a time and abstract from information acquisition, they cannot explain why parents avoid cheap information that would resolve their uncertainty (F2); a parent who retrieves unfavorable memories can still recognize that her reading of the program is uncertain. Nor do they explain why a change in cues leads to sustained engagement rather than a temporary shift in judgment (F3).

7.3 Preferred interpretation: from substitutes to complements

The interpretation I favor is that parental engagement shifts parents from viewing agriculture and education as substitutes to viewing them as complements. Without engagement, parents interpret the program through a familiar, precise, and widely shared default view. This view rests on several preconceptions: that school gardens are drudgery that trains obedient farm assistants rather than building agribusiness skills; that farming is what children fall back on when schooling fails, so that education is the route out of agriculture; and that

hours spent on a farm are hours taken away from learning. Under this view, a school that adds a farm is a school that teaches less. Children's reports that school time is spent farming confirm the view (F1, F5), engaging with the program appears pointless (F2), and the natural response extends to schooling itself, so that studying hours and attendance fall (F4). This is better described as an aspirations failure than as a pure information friction of the kind studied by Jensen (2010), Bergman (2021), and Dizon-Ross (2019). In those settings, parents lack important facts about returns to education or their children's performance. In this setting, parents have access to the relevant facts about SBAE, but they lack an interpretation of the program under which these facts are valuable.

The alternative interpretation treats the school farm as applied science. Under this interpretation, farm management skills raise the returns to both schooling and farming, and students' home projects test the promoted techniques on the household's own plots. This interpretation is less precise than the default view, for a good reason: it depends on several conditions that a parent cannot verify from what she observes. For SBAE to pay off, the program must be implemented well, its funding must continue after donor support ends, the techniques must suit local soils, students must acquire management skills rather than simply perform tasks, and teachers must change how they teach. As a result, new facts that are consistent with the default view do not lead parents to abandon it.

In a companion theory paper, I develop a framework in which the engagement video works by contrasting SBAE with what parents already know, rather than by supplying new information. By showing how SBAE differs from typical school gardens, the video challenges a default view that is shared across the community and makes it harder to maintain, so that parents begin to give weight to the complements interpretation. Holding two interpretations that recommend different actions creates uncertainty about which interpretation is correct, and with it curiosity. Parents then look for evidence on the points where the two interpretations disagree, for instance on what students are actually able to do. When the complements interpretation is correct, this search reinforces the shift over time. This explains why a low-content trigger can produce a lasting response (F3), and why parents who had previously avoided information begin to seek it out. Without engagement, parents hold only the default interpretation and have no reason to question it, so they see little value in visiting the school.

The contrast between the two engagement treatments (Section 6.5) is consistent with this interpretation. The annual farmer field day provides more technical content than the video. Yet only the video activates students' home projects and intra-household diffusion, raises parents' technology adoption, and reduces dropouts. Evidence from my companion paper (JMP2, Lee, 2025) is also consistent with this interpretation. 97% of parents who watch the video anticipate growth in students' farm management abilities in one year, including their cassava yields, knowledge about farm management, and knowledge about promoted techniques. Their beliefs about students' cassava yields remain significantly more optimistic at both midline and endline compared to parents in schools without video sessions. Moreover, parents' beliefs about the difference between students' cassava yields and those of typical adult farmers — measured immediately after watching the video — strongly predict their midline responses to SBAE. One standard-deviation increase in these beliefs is associated with a 14% increase in visits to school farms, a 22% increase in the likelihood of observing school farm harvests, a 31% increase in the likelihood of working on school farms, and a 12% increase in the likelihood of meeting with the responsible teacher. These associations do not merely reflect social desirability bias, since one standard-deviation increase is also associated with a 2.7 percentage point increase in parents correctly naming field officers (from a base of 11%) and 9% higher attendance at parent-teacher meetings.

8 Concluding remarks

Finding an effective delivery model for agricultural extension stands out as one of the foremost challenges in African rural development. Using a randomized experiment with 197 schools, this paper provides proof of

concept that schools can be a promising entry point for diffusing yield-improving agricultural technologies. This promise lies in two parts. First, I show that — contrary to concerns about pitfalls — with appropriate implementation, school-based agricultural extension can in fact encourage school attendance alongside increasing households' technology adoption. Second, I show that due to scale economies in leveraging the school infrastructure, the program offers both scope of impact and cost-effectiveness. Specifically, I show that the program increases households' technology adoption, students' savings, and school attendance. It is on par with the best village-based extension programs in increasing technology adoption; more cost-effective than non-school-based agricultural training in increasing youth's savings; and more cost-effective than conditional cash transfers in increasing school attendance.

Alongside the promise of school-based agricultural extension, it should be noted that its large, positive effects are achieved only with parental engagement and sub-treatments discussed in Lee (2025). To the extent that school-based development programs have strategic advantages in logistics (e.g. scalability and sustainability), this result offers two general lessons. First, we should conduct joint evaluations of school-based development programs and household interventions, as this paper reveals that the former could have hidden potentials that are only unlocked with the latter. Second, we should investigate ways to enhance students' agency in amplifying the impact of school programs, not just as a matter of youth empowerment, but as ways to increase the cost-effectiveness of school programs.

Beyond immediate impacts, this paper demonstrates that schools can serve as sustainable rural development hubs when coupled with low-cost parental engagement events. The first step is to motivate students to attend schools, triggering a virtuous cycle in which schools are strengthened through improved incomes and provide higher-quality education. However, while the current RCT demonstrates positive effects on technology adoption and school attendance, it shows no improvements in students' academic performance or agribusiness knowledge. The second step, therefore, is to synergize SBAE with complementary interventions. One direction that I am taking is to integrate the highly effective, student-centered "Learning to Learn (LTL)" pedagogy into SBAE, which is shown in a randomized trial in Uganda by Nourani et al. (2025) to be amongst the top 5% education interventions in terms of learning-adjusted years of schooling. While complementary with the student-centered, experiential instruction promoted by SBAE, this intervention does not prescribe fixed teaching methods. Instead LTL prompts teachers to reflect on how they learn, emphasizing active learning, critical thinking, and problem-solving skills essential for students to effectively apply agricultural knowledge in real-world contexts. Another direction is to link SBAE to home-grown school-feeding programs, creating direct market demand for students' agricultural produce while expanding the scope of agribusiness education. A third direction is to leverage the high-frequency student data collection system developed during the RCT to conduct iterated randomized trials on different agricultural technologies. This approach is particularly promising when combined with algorithmic farming advice, a partnership currently under development.

As of writing, the current randomized trial has significant policy impact in sub-Saharan Africa, scaling to all rural schools in Liberia by 2035, with expansion to Côte d'Ivoire and Kenya beginning in 2026. In the long term, this holistic approach of SBAE can contribute to our understanding of macroeconomic development in agrarian economies. First, it addresses whether schools can simultaneously promote rural livelihoods while equipping students with adaptable skills for structural transformation toward higher-value sectors. Second, it contributes to our ability to understand and shape the direction of technical change through a system of high-frequency feedback and enhanced data collection.

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A Attrition

Appendix Figure A1 traces the flow of schools, students, and parents from allocation to analysis, following the CONSORT guideline for cluster-randomized trials. Appendix Table A1 reports attrition rates of students (midline and endline) and parents (endline) by treatment arm, and the reasons for attrition. Section 3.5 discusses the patterns; Online Appendix Table OA3 shows that selection into the surveyed samples does not differ across arms.

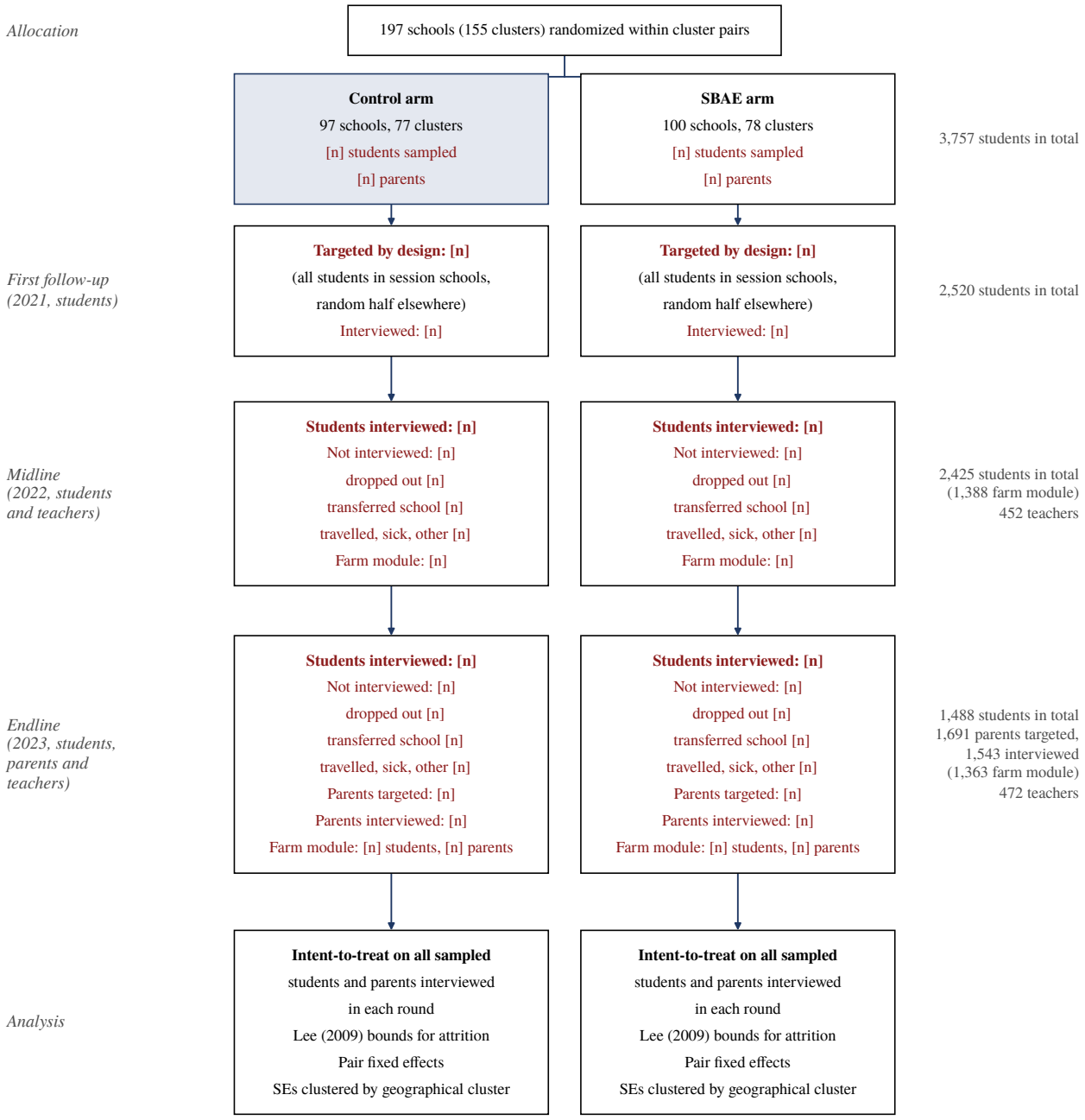


FIGURE A1. PARTICIPANT FLOW

Notes: This figure follows the CONSORT reporting guideline for cluster-randomized trials. Allocation was at the cluster level within matched pairs; all 197 schools remained in their assigned arm and no cluster was lost. Student counts refer to the 3,757 students sampled at baseline (Section 3.3); "not interviewed" lists the reasons recorded by the survey teams, which are the attrition categories in Appendix Table A1. The first follow-up targeted all sampled students in the 50 session schools and a random half elsewhere, so its non-interviews are by design and not attrition. Parents were surveyed at endline only, in a random 45 percent of households stratified by school. Teachers are repeated cross-sections and are not followed. Bracketed entries are placeholders to be filled from the tracking data.

TABLE A1—BALANCE IN ATTRITION RATES ACROSS TREATMENT ARMS

Panel A: Balance in attrition			
	Midline Students Attrited (0/1)	Endline Students Attrited (0/1)	Endline Parents Attrited (0/1)
$SBAE_p$	0.020	0.028	0.042**
Standard errors	(0.035)	(0.034)	(0.020)
$SBAE_p \times Session_s$	-0.073**	-0.092***	0.017
Standard errors	(0.034)	(0.030)	(0.020)
$SBAE_p \times FFD_s$	0.001	0.046	-0.040**
Standard errors	(0.036)	(0.037)	(0.020)
Control Mean	0.36	0.48	0.09
Overall Mean	0.35	0.48	0.12
Observations	3757	3757	1689
Number of clusters	155	155	155

Panel B: Reasons for attrition			
	Dropped out or switched schools	Dropped out or switched schools	Travelled
	34.9%	45.5%	4.6%
	— Confirmed dropped out	— Confirmed dropped out	Relocated
	18.3%	23.9%	4.3%
	— Confirmed switched schools	— Confirmed switched schools	Working
	16.3%	20.9%	1.1%
	Travelled	Travelled	Sick
	0.2%	0.6%	0.9%
	Moved to foreign countries	Moved to foreign countries	Moved to foreign countries
	0.1%	0.5%	0.2%
	Passed away	Passed away	Passed away
	0.1%	0.2%	0.9%
	Other reasons	Other reasons	Other reasons
	0.4%	0.9%	0.4%

Panel C: Balance in major attrition reasons			
	Dropped out or switched schools (0/1)	Dropped out or switched schools (0/1)	Travelled or relocated (0/1)
$SBAE_p$	-0.000	0.024	0.038*
Standard errors	(0.033)	(0.033)	(0.021)
$SBAE_p \times Session_s$	-0.061*	-0.099***	0.014
Standard errors	(0.034)	(0.031)	(0.023)
$SBAE_p \times FFD_s$	0.014	0.046	-0.047**
Standard errors	(0.035)	(0.037)	(0.020)

Notes: This table shows a test of balance for the attrition rates among students and parents in the midline and endline surveys. Panel A regresses a binary indicator of attrition on a binary indicator of the SBAE treatment, an interaction term between the SBAE treatment and the program promotional session treatment, and an interaction term between the SBAE treatment and the annual farmer field day treatment. Panel B shows the distributions of reasons for attrition in each survey. Panel C repeats the balance test in Panel A, but using only attrition due to major reasons. All regressions include pair-of-clusters-of-schools fixed effects and a binary indicator of faith-based schools. Standard errors are clustered at the cluster-of-schools level.

B Cost-Benefit and Cost-Effectiveness Analyses

This appendix details the assumptions and results summarized in Section 5.

Assumptions.

Appendix Table B1 lists all key assumptions in the cost-benefit and cost-effectiveness analyses. I first define the program beneficiaries. Corresponding to the randomized trial, I consider a three-year period in which three cohorts of students in 100 SBAE schools are targeted. Since sampled students represent a random 30% of students in targeted grades, the program reaches roughly 65 students per school in each cohort—a total of 19,500 students across all cohorts and schools. For the analysis of technology adoption, which may include spillovers of farming practices to non-parent households, it is important to note that approximately 410,000 people reside within 2 kilometers of the SBAE schools (Liberia Institute of Statistics and Geo-Information Services, 2008).

The total cost over this three-year period is $801,139 (2023 US dollars), or about $40 per student. This covers two components: program implementation ($621,139), including training extension officers and teachers, monthly school monitoring, setting up school farms, and hosting national events for extracurricular clubs; and parental engagement ($180,000), including a promotional session for one parent per student and an annual farmer field day.

An important consideration for program sustainability is the potential for drastically lower costs after the initial three years. Setup costs are not repeated, and I assume monitoring visits are sharply curtailed to once a year, reducing three-year implementation costs from $621,139 to $120,000—a 63% drop in total costs.

Turning to the side of program benefits, the first important component is the increase in students' savings — Section 4.2 shows that in the second year of the program, there is a 21% increase in annual savings from a base level of $20 (2022 US dollars) for students who have not dropped out from schools. I assume that this increase carries over to the third year.

The second important component is the increase in students' school attendance — due to both reduced dropouts and reduced absenteeism. To combine effects on both dropouts and absenteeism, I focus on the treatment effect on the imputed total number of days in attendance, which is 17.6 according to Online Appendix Table OA6.

For the cost-benefit analysis, we must monetize this increase in school attendance. Adopting a conservative approach, we assume an 8% annual return to schooling, implying a return of 0.04% per school day (based on approximately 200 school days per year).[19] Applying a 10% discount rate and assuming that students work for 30 years, we estimate that each additional school day generates $1.89 in discounted lifetime earnings.

The third component, which is also the most uncertain component, is parents' benefits from technology adoption. To calculate these benefits as conservatively as possible, I consider only adoption of bed preparation in one season (only in verified third year) and in 1/4 of the typical cultivated area of roots and vegetables. This tends to understate the benefits as (i) SBAE also increases joint adoption of multiple techniques (as we discuss in Section 4.2.1) tend further raise yields; (ii) parents who experience improved yields tend to continue using the technologies. Further, as yields in the general population are often found to be lower than in agronomic trials, I assume only half of the yield gains in agronomic trials. I use cassava and sweet potatoes as the representative crops (each comprising half of the area of adoption) — these crops are widely grown by Liberian farmers as their flexible use helps satisfy subsistence needs, but have lower prices in the market than

[19] The global average return to an additional year of schooling is approximately 8.8% (Psacharopoulos and Patrinos, 2018). We use 8% as a conservative estimate. Returns in sub-Saharan Africa, especially at the secondary level, are typically higher, ranging from 10–13% (Montenegro and Patrinos, 2014). A World Bank document (World Bank, 2019) also reports that based on household survey data in 2016, individuals who completed senior high school earned on average 18 percent more than those who completed only junior high school, though this descriptive finding reflects correlation rather than causal returns. Downward revision from these optimistic estimates is reasonable, as standard Mincerian estimates may overstate returns for rural populations who are more likely to remain in the agricultural or informal sector, where credential-based wage premiums are less pronounced.

other root crops and vegetables. I also consider the potential increase in labor costs due to constructing beds — 150 hours per hectare. Taking all factors into account, I calculate that each adoption of bed preparation creates a net production value of $64.6.

Cost-benefit analysis.

We distinguish between two types of analysis: first, whether benefits justify costs in the first three years, with known data and the assumptions above; second, whether SBAE is a good long-term investment, viewing from the perspective of a policymaker who commits a stream of spending on SBAE.

Appendix Figure B1 presents the benefits and costs over the first three years, revealing two key observations. First, benefits exceed costs during this period, yielding a benefit-cost ratio of 1.33. Second, even when excluding benefits from parents' technology adoption—a conservative approach for those skeptical about the magnitude of yield gains—the benefit-cost ratio remains at 0.98. In other words, the benefits accruing to students alone nearly justify the program, even under the most pessimistic assumptions regarding parents' gains from technology adoption and potential spillovers of improved farming practices to non-parent farmers.

Appendix Figure B1 demonstrates that the augmented SBAE program is economically justified in the short term. However, this analysis overlooks a key strength of SBAE: its emphasis on capacity building means the program can be sustained through periodic teacher training alone, without ongoing monitoring visits from extension officers. This raises a critical question: what are the long-term returns on SBAE? The answer hinges on two competing considerations. On the one hand, costs drop substantially after the initial three years—setup costs are not repeated, and monitoring visits are curtailed to once a year while teacher training is retained, reducing implementation costs from $621,139 to $120,000, a 63% reduction (inclusive of parental engagement). On the other hand, the key uncertainty is whether the program remains equally effective once external support is scaled back.

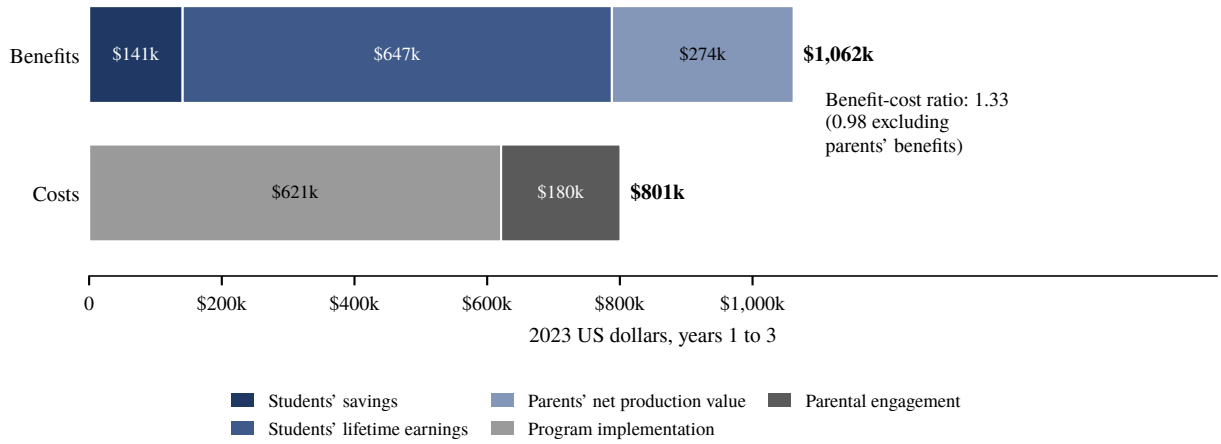


FIGURE B1. BENEFITS AND COSTS IN THE FIRST 3 YEARS OF THE AUGMENTED SBAE PROGRAM

Notes: This figure plots all components of benefits and costs of the augmented SBAE program in the three-year RCT period, in thousands of 2023 US dollars. Students' lifetime earnings are discounted to present value. The benefit-cost ratio divides total benefits by total costs; excluding parents' net production value, it is 0.98. Assumptions are listed in Appendix Table B1.

Appendix Figure B2 presents the 10-year cost-benefit analysis under two scenarios. During the initial three-year implementation period, the benefit-cost ratio is 1.33. In Panel A, which assumes program benefits remain unchanged after the first three years, the benefit-cost ratio for Years 4–10 rises to 3.54, yielding a cumulative benefit-cost ratio of 2.36 over the ten-year horizon. In Panel B, which assumes a more conservative scenario where program benefits decline to 50% after the first three years, the benefit-cost ratio for Years 4–

TABLE B1—KEY ASSUMPTIONS AND PARAMETER VALUES

Item	Value	Justification
Scope & reach		
Period of consideration	3 years	RCT period
Number of schools	100	Schools in SBAE arm
Direct beneficiaries (households)	19,500	3 cohorts × 6,500 students
Community population (within 2 km of schools)	410,000	2008 census data
Cost assumptions (Years 1–3)		
Total cost	\$801,139	
Program cost	\$621,139	Training of extension officers and teachers, monitoring costs, setup costs
Parental engagement cost	\$180,000	Program promotional session to every cohort and annual farmer field day
Cost assumptions (Years 4–6 and beyond)		
Total cost	\$300,000	
Program cost	\$120,000	Dropping setup costs; monitoring visits curtailed to once a year
Parental engagement cost	\$180,000	Same as before
Students' savings		
Base annual savings per student	\$21	Average in control schools (translated into 2023 US dollars)
Growth rate of savings	21%	RCT finding on Year 2: Section 4.2.3
Savings increase (Year 2) for non-dropouts	4.41	92% increase
Savings increase (Year 3) for non-dropouts	4.41	Assuming saving increase in Year 2 carries over
Students' discounted lifetime earnings		
Number of additional school days per student	17.6	RCT finding: Section 4.2.2
Base annual income	\$500	Typical annual income in rural Liberia
Return on additional school day	8/200%	Implying annualized return of 8%
Discount rate	10%	Standard rate in program evaluation
Present value per extra school day	\$1.89	Working for 30 years after finishing school
Parents' benefits from adopting bed preparation		
Number of parents adopting	4,251	RCT finding: 19,500 households × 2 parents × 10.9%
Cassava base yield (kg/ha)	8,280	FAOSTAT database
Cassava yield increase	15%	1/2 of the 30% yield gain in agronomic trial
Sweet potato base yield (kg/ha)	1,283.4	FAOSTAT database
Sweet potato yield increase	35%	1/2 of the 70% yield gain in agronomic trial
Adoption area per household (ha)	0.15	1/4 of typical cultivated area for roots and vegetables
Labor increase (hrs/ha)	150	Estimates from FAO guidebooks
Cassava price (USD/kg)	\$0.21	Average market price in 2023 from FAO
Sweet potato price (USD/kg)	\$0.30	Average market price in 2023 from FAO
Labor cost (USD/hour)	\$0.43	Agricultural wage rate in rural Liberia
Net increase in production value per adoption	\$64.60	Increased production value net of increased labor cost

Notes: This table lists all essential assumptions used for cost-benefit and cost-effectiveness analyses in Section 5 and this appendix. All monetary values are expressed in 2023 US dollars.

10 is 1.77, with a cumulative benefit-cost ratio of 1.53 over the ten-year period. While Panel A highlights SBAE's potential to deliver even higher returns over time, the results in Panel B provide a robustness check that strongly justifies long-term policy commitment. By capitalizing on the institutional capacity and teacher training established during the initial implementation period, sustained returns can be achieved at a fraction of the initial costs.

Cost-effectiveness analysis.

We now compare SBAE's cost-effectiveness in addressing the dual challenges of diffusing agricultural technologies and increasing school attendance against major alternatives in sub-Saharan Africa, drawing on randomized trial evidence from Innovations for Poverty Action. The Online Appendix provides further details on these programs and cost-effectiveness calculations.

Appendix Figure B3 compares the impact per $100 of the augmented SBAE program with other African programs evaluated using randomized trials. I first assess SBAE's cost-effectiveness in increasing technology adoption by calculating the number of households that adopt promoted techniques per $100 invested. For SBAE, I measure adoption of bed preparation on parents' parcels at endline. A key unknown—which I will track in a 2026 follow-up survey—is the adoption rate among randomly sampled non-parent farmers in nearby communities. I assume this rate (for communities within 2 kilometers of schools) ranges from 0% to 5%, in line with estimates of positive spillovers in farming practices from the social learning literature.

The prevailing mode of agricultural extension in developing countries is the contact farmer model, in which extension agents train selected village farmers who are then expected to disseminate practices to their peers. As Suri and Udry (2022) summarize, this approach has generally proven ineffective at triggering adoption beyond direct trainees. Efforts to improve extension have largely focused on enhancing this existing model—by tweaking how contact farmers are trained, leveraging information technologies to deliver messages to farmers, or increasing social learning by choosing contact farmers more carefully. I compare SBAE, which represents a departure from the contact farmer approach, against these RCT-evaluated programs. Despite differences in the specific techniques promoted, all programs target soil management—a pervasive agronomic problem—making them reasonably comparable.

Appendix Figure B3 shows that, with the exception of BenYishay and Mobarak (2019), SBAE's impact per $100—0.46 to 0.94 households adopting bed preparation, or 0.55 to 1.02 households adopting bed preparation *and* a complementary soil management technique—far exceeds that of extension programs using the contact farmer model. Kondylis et al. (2017) attempt to address imperfect implementation and accountability among extension agents by adding a centralized training workshop for contact farmers. While this increases adoption among contact farmers, it fails to reach untrained farmers, yielding an impact of just 0.11 households adopting pit planting per $100. Two other notable studies evaluate the use of farmer field days and video messaging. These interventions are similar in cost to the annual farmer field day and promotional video I introduce as parental engagement treatments in SBAE, but differ in that agricultural training targets adults rather than students. Fabregas et al. (2023) study the effects of farmer field days in promoting Mavuno fertilizer adoption. With an intent-to-treat effect of 4 percentage points and 100 to 300 farmers reached per field day, the impact is 0.15 to 0.46 households per $100. Van Campenhout et al. (2021) evaluate video-mediated extension messages designed to teach optimal crop spacing to maize farmers in Uganda. This technique is comparable to bed preparation in the present study, as both are relatively easy to adopt. Despite encouraging effects on knowledge, practices, and yields, the program induces only 0.62 households to adopt per $100 spent, placing it at the lower end of SBAE's cost-effectiveness range. The only program that outperforms SBAE is village-based extension in Malawi augmented with performance incentives for contact farmers (BenYishay and Mobarak, 2019)—and only when the outcome is composting adoption (1.53 households per $100). When considering pit planting, a technique more comparable to bed preparation given similar labor intensity and complexity, the impact falls to 0.6 households per $100, within the lower half of SBAE's range.

Second, I assess SBAE's cost-effectiveness in increasing school attendance by calculating additional

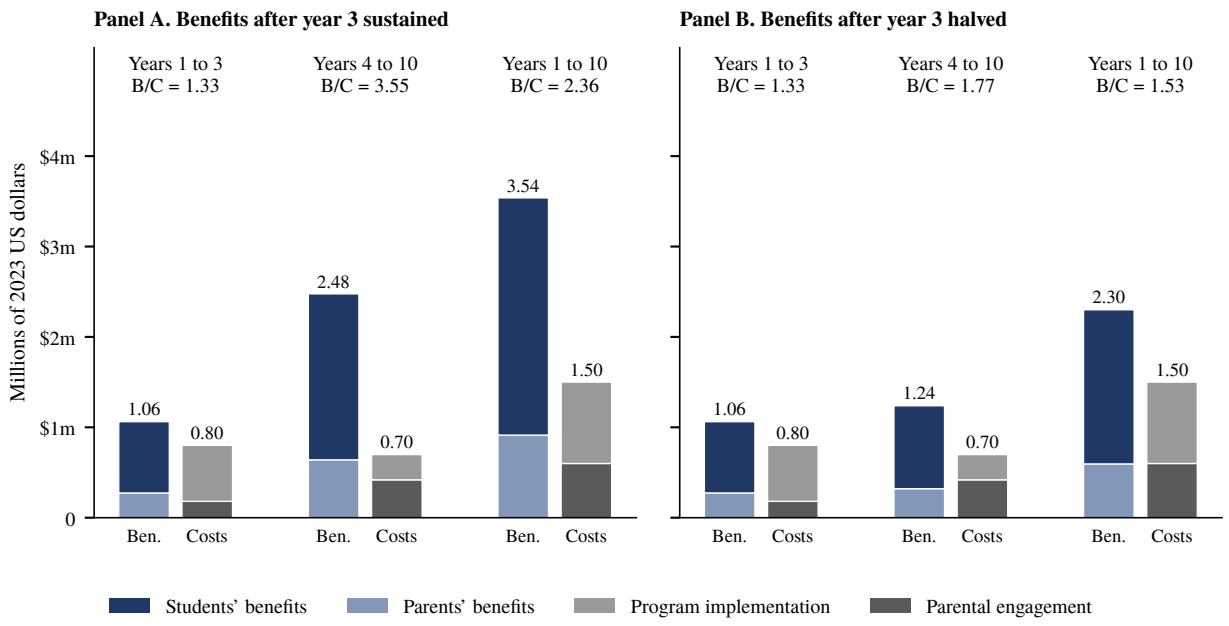


FIGURE B2. SBAE'S BENEFIT-COST RATIOS UNDER DIFFERENT ASSUMPTIONS ABOUT POST-RCT PROGRAM EFFECTIVENESS

Notes: Years 1–3 are as in Appendix Figure B1. For each of Years 4–10, benefits equal the average annual benefit in Years 1–3 (Panel A) or half of it (Panel B), and costs are $40,000 for program implementation (teacher training and annual monitoring visits) and $60,000 for parental engagement. Numbers above the bars are totals in millions of 2023 US dollars; amounts across years are summed without further discounting. B/C is the benefit-cost ratio for each period.

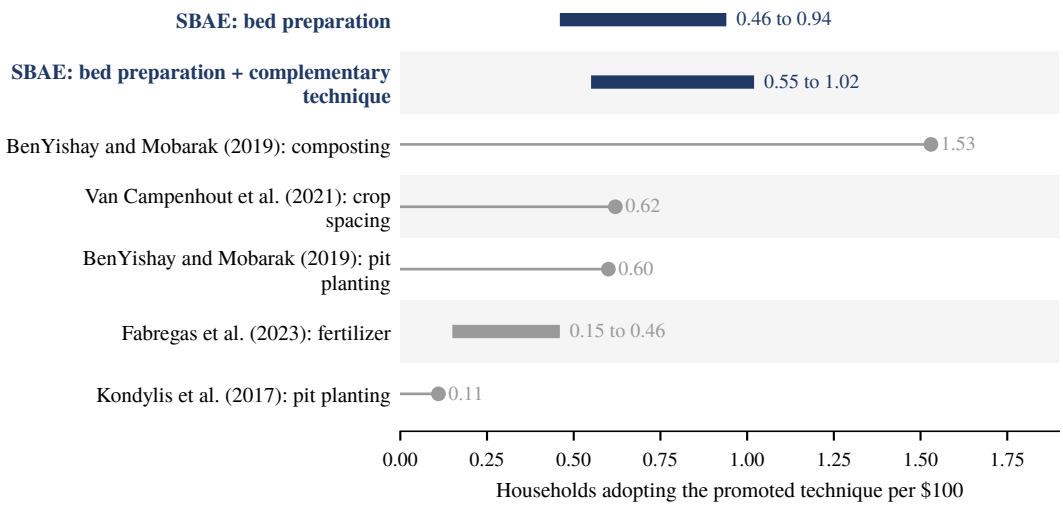
school days per student per $100 invested. I benchmark against conditional cash transfers (CCTs), one of the most popular approaches to boosting school attendance in sub-Saharan Africa and among the most rigorously evaluated. Drawing on Baird et al. (2011)'s RCT in Malawi, Appendix Figure B3 shows that the augmented SBAE program (43.46 days per $100) is 1.7 to 2.5 times as cost-effective as CCTs (17.06–24.98 days per $100). My companion paper (JMP2, Lee, 2025) shows that this advantage widens further when SBAE is augmented with both parental engagement and information nudges signaling parents' openness to delegate farms to students: as students shift toward sales-oriented crops to finance school fees, cost-effectiveness rises to 59.51 days per $100—2.4 to 3.5 times that of CCTs.

Generalizability.

How generalizable are these results? I speculate that there would be a shift of balance between short-term and long-term benefits as SBAE is extended to countries that are higher up the ladder of economic development. The short-term benefits might become smaller for three reasons. First, there is less room to promote agricultural technologies that involve low monetary costs, can easily be used by students, and yet improve yields by a large extent. This would result in less dramatic increases in students' savings. Second, there is less room to improve school attendance as the dropout rates are much lower. Third, informational barriers are less likely to be the main constraints for parents' adoption of off-the-shelf agricultural technologies.

In the long term, however, SBAE's framework could generate more benefits for countries higher up the economic ladder, especially if SBAE could partner or integrate with complementary programs. First, the returns from promoting more advanced techniques, such as improved crop varieties and proper use of chemical fertilizers, would far exceed the returns from the promoted techniques in Liberia. While factors such as credit constraints would limit the use of these techniques, the SBAE framework remains powerful in alleviating

Panel A. Technology adoption



Panel B. School attendance

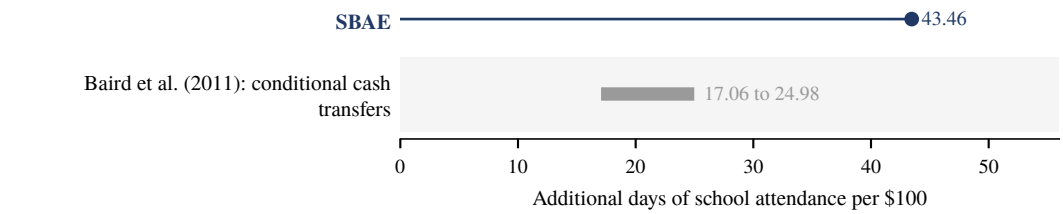


FIGURE B3. COMPARING THE AUGMENTED SBAE PROGRAM WITH ALTERNATIVES USING RANDOMIZED TRIAL EVIDENCE

Notes: This figure plots the cost-effectiveness of the augmented SBAE program (dark blue) and alternative programs evaluated in randomized trials (gray), in terms of impact per $100 invested. Panel A counts households adopting the main promoted technique; SBAE counts adoption of bed preparation on parents' parcels at endline (alone, or with a complementary soil management technique), and its range reflects assumed adoption of 0% to 5% among non-parent farmers within 2 kilometers of schools. The range for Fabregas et al. (2023) reflects 100 to 300 farmers reached per field day. Panel B counts additional days of school attendance; the range for conditional cash transfers is from Baird et al. (2011). Bars show ranges and dots single estimates. All monetary values are expressed in 2023 US dollars. Relevant calculations can be found in the Online Appendix.

information barriers about the extent of yield gains. Second, rather than encouraging school attendance per se, education systems in these countries are more well-equipped to deliver the potential of student-centered learning under SBAE. In the very first step, SBAE could provide an entry point to train and motivate teachers to use more effective pedagogies in teaching science outside the few structured lessons related to agriculture, thereby improving students' understanding of and performance in science and related subjects. More importantly, the development of an educational culture that is centered on scientific understanding and open inquiries would facilitate bottom-up experimentation of farming techniques by students and their households, thereby sustaining the SBAE program without external extension support.